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9.1 INTRODUCTION

9.1.1 This chapter has been prepared by FPCR Environment & Design Ltd and assesses the likely significant effects of the Proposed Development in terms of ecology and nature conservation and is based upon an ecological assessment of desk study information and habitat and species surveys.

9.1.2 The chapter describes the assessment methodology; establishes the baseline conditions currently existing at the Application Site and surroundings; the likely significant environmental effects; the mitigation measures required to prevent, reduce or offset any significant adverse effects; and the likely residual effects after these measures have been employed.

9.1.3 This chapter should be read in conjunction with **Appendices 9.1** to **9.8**, which provide further detail regarding habitat and species survey and assessment.

9.2 ASSESSMENT METHODOLOGY

Scope

9.2.1 The Application Site (hereafter referred to as the Site) extends to approximately 162 hectares and lies adjacent to the eastern edge of Swindon. Its southern boundary follows the main Bristol to London Mainline railway, and/or the A420 also to the south. The northern boundary is formed, in part, by South Marston in its western extent and, in part, by Nightingale Lane and Wood in its eastern extent. The River Cole lies to the east.

9.2.2 Historically, a wider Study Area was surveyed including most of the area covered by *Policy NC3: New Eastern Villages - including Rowborough and South Marston Village Expansion* of the *Swindon Borough Local Plan 2026: Submission Document* (June 2013) and identified in the Proposals Map contained within that document. Some of these surveys are now outdated. Nevertheless, where relevant information exists it has been used to inform the scope of surveys and the assessment of potential effects.

9.2.3 This information concerning the ecological interest of the Site and its wider context includes:

- Earlsgate EIA: Chapter 4 Ecology (Pegasus Planning Group, 1999), which covered land outside the Application Site to the south of the A420;

- Ecological Appraisal, New Marston (FPCR 2002), which broadly covered the existing Study Area in addition to the curtilage of South Marston Farm; and

- Ecology Scoping Report (FPCR 2007), which included much of the Policy NC3 New Eastern Villages area.

Data sources

9.2.4 In order to compile existing baseline information, relevant ecological information was requested from a range of organisations/individuals for the purposes of this assessment:

- Wiltshire and Swindon Biological Records Centre;

- Wiltshire Wildlife Trust;
- Natural England Invertebrate Site Register (ISR);
- Botanical Society of the British Isles (BSBI) recorder for Wiltshire;
- Wiltshire Badger Group;
- Wiltshire Bat Group;
- Environment Agency;
- North Wiltshire RSPB Group; and
- County Reptile and Amphibian Recorder.

9.2.5 A number of on-line resources were also searched for relevant records including:

- the Multi Agency Geographic Information for the Countryside (MAGIC) website[1]; and
- the National Biodiversity Network (NBN) website[2] (for contextual information only - no data has been used in the compilation of this chapter).

9.2.6 Further inspection, using colour 1:25,000 OS base maps (www.ordnancesurvey.co.uk) and aerial photographs from Google Earth (www.maps.google.co.uk), was also undertaken in order to provide additional context and identify any features of potential importance for nature conservation in the wider countryside.

9.2.7 The search area for biodiversity information was related to the significance of sites and species and potential zones of influence are described in Table 9.1.

Table 9.1: Search Area

Distance	Source of Information
10km	Sites of International Importance, e.g. Special Areas of Conservation (SAC), Special Protection Area (SPA), Ramsar Site
2km	Sites of National or Regional Importance, e.g. Sites of Special Scientific Interest (SSSI) and Sites of County Importance, i.e. Sites of Importance for Nature Conservation (SINCs)
1km	Species records, e.g. protected, Species of Principal Importance, Wiltshire BAP or notable species

[1] http://magic.defra.gov.uk/
[2] http://data.nbn.org.uk/

Flora

9.2.8 The Study Area encompassed the Site and some other areas of adjacent habitat as indicated on the Phase 1 Habitat Plan (**Appendix 9.1**).

9.2.9 Survey methods followed the extended Phase 1 Survey technique as recommended by Natural England[3]. This involved a systematic walk over of the Study Area by an experienced ecologist to classify the broad habitat types and to particularly identify any habitats of principal importance for the conservation of biodiversity as listed within Section 41 (S41) of the Natural Environment and Rural Communities (NERC) Act (2006).

9.2.10 Species lists were compiled for representative habitats. Suitably qualified and experienced botanists, who are members of the Institute of Ecology and Environmental Management (IEEM), undertook all botanical survey work.

9.2.11 Hedgerows were surveyed using the Hedgerow Evaluation and Grading System (HEGS)[4]. This method of assessment includes noting canopy species composition, associated ground flora and climbers, structure of the hedgerow including height, width and gaps, associated features including number and species of mature trees, banks, ditches and grass verges.

9.2.12 Each hedgerow is given a grade using HEGS with the suffixes '+' and '-', representing the upper and lower limits of each grade respectively. These grades represent a continuum on a scale from 1+ (the highest score and denoting hedges of the greatest nature conservation priority) to 4- (representing the lowest score and hedges of the least nature conservation priority) as follows:

Grade 1 – High to very high value
Grade 2 – Moderately high to high value
Grade 3 – Moderate value
Grade 4 – Low value

[3] JNCC. (1990). Handbook for Phase 1 habitat survey – a technique for environmental audit. Peterborough: JNCC
[4] Clements, D.K., & Tofts, R.J.(1992). Hedgerow Evaluation and Grading System (HEGS): A methodology for the ecological survey, evaluation and grading of hedgerows. Countryside Planning and Management

9.2.13 Hedgerows graded 1 or 2 are considered to be a priority for nature conservation.

9.2.14 The hedgerows were also assessed against the Wildlife and Landscape criteria contained within Statutory Instrument No: 1160 – The Hedgerow Regulations 1997[5] to determine whether they qualified as 'Important Hedgerows' under the Regulations. This was achieved using a methodology in accordance with both the Regulations and DEFRA[6] guidance.

9.2.15 Mature trees within the Site were assessed for their status as veteran trees using DEFRA[7] and Natural England guidance[8].

9.2.16 Where areas of relatively homogenous vegetation occurred further detailed botanical survey was undertaken in order to provide information on the relative cover of species.

Fauna

9.2.17 Detailed survey methodologies for flora and fauna are provided in the species specific reports in **Appendices 9.1** to **9.8**. A summary of the surveys undertaken is provided in Table 9.2 below.

Table 9.2: Faunal Survey Timetable

Species/Taxa	Survey	Date
Great Crested Newt	Presence/absence survey as described in the Great Crested Newt Mitigation Guidelines (English Nature 2001)	Various surveys completed in 2002, & 2006,2009 and most recently in 2012 on the: 25th April 09th May 14th May

[5] http://www.legislation.gov.uk/uksi/1997/1160/contents/made
[6] DEFRA. (1997). The Hedgerow Regulations 1997. A Guide to the Law and Good Practice. London: HMSO
[7] Rural Development Service. (2006). Environmental Stewardship-Farm Environment Plan Guidance 009. Available on line at http://www.naturalengland.gov.uk/Images/fepveterantree_tcm6-6492.pdf
[8] Natural England. (1999). Veteran Trees –A Guide to Good Management. Available online from http://publications.naturalengland.org.uk/publication/75035

Species/Taxa	Survey	Date
		27th May
Bats	Activity Surveys comprising transect surveys in each month from April to October based on the methods outlined in the Bat Surveys Good Practice Guidelines, 2nd Edition (L Hundt 2012) Automated Static Bat Detector Survey up to 3 static detectors deployed from April to October in strategic locations within Site. Assessment of mature trees for bat roost potential	Completed summer 2006 and 2008 Updated Spring to Autumn 2013 Completed 10th and 30th April 2013
Reptiles	Presence/absence survey following survey protocol outlined in the Herpetofauna Workers Manual (Gent and Gibson, 1998) and the Froglife Advice Sheet 10 - Reptile Survey (Froglife 1999).	Undertaken from April 2013
Birds	Breeding bird survey. Three visits to estimate species richness and territories held. Winter Bird Survey. Visits conducted in each month from December to February following WeBS methodology	Completed in 2006 and subsequently on the 12th May, 1st June and 21st June 2011 Completed in 2006 and subsequently in 2012/13
Badgers	Survey of all habitats within the Site for signs of use or occupation following methods of Cresswell, Harris and Jeffries.	Completed 2006 and May 2012
White-clawed crayfish	Presence/absence survey. Methods followed Peay (2003) and included torchlight survey and refugia search where possible.	Completed October 2006 and updated August 2013
Otter and Water Vole	Presence/absence survey following methods outlined in Strachan and Moorhouse (2010)	Completed in 2006 and 2010

Assessment Approach

Methodology

9.2.18 Reference has been made to the Guidelines for Baseline Ecological Assessment (1995) and to the Institute of Ecology and Environmental Management (IEEM) Guidelines for Ecological Impact Assessment (EcIA) in the United Kingdom (July 2006). These guidelines aim to give a degree of consistency in approach to evaluating the importance of the ecological features within the Site and any effects or impacts a scheme will have upon them.

9.2.19 The activities associated with the construction and occupation of the Proposed Development have been identified, together with the likely range within which their influence will be felt, given the nature of the area. The ecological features, which may be affected by such activities, have been evaluated within a geographical framework which is based on the ecological status of the features, but which also reflects a wide range of legislation and governmental guidance.

Assessment Evaluation

9.2.20 An assessment of the nature conservation value of the Application Site (sensitivity) was made following the criteria suggested in IEEM (2006) as follows; International, National, Regional, County, District and Local. A summary is also provided in Table 9.3.

Table 9.3: Evaluation of Nature Conservation Importance

Value/ sensitivity	Receptors
International/High	An internationally designated site or candidate site (SPA, pSPA, SAC, cSAC, pSAC , Ramsar site, Biogenetic Reserve) or an area which meets the published selection criteria for such designation, irrespective of whether or not it has yet been notified. A viable area of a habitat type listed in Annex I of the Habitats Directive or smaller areas of such habitat which are essential to maintain the viability of a larger whole. Any regularly occurring population of an internationally important species, which is threatened or rare in the UK (i.e. it is a UK Red Data Book species or listed as occurring in 15 or fewer 10km squares in the UK, of uncertain conservation status or of global conservation concern. A regularly occurring, nationally significant population/number of any internationally important species.
National/high	A nationally designated site (SSSI, NNR, Marine Nature Reserve) or a discrete area, which meets the published selection criteria for national designation (e.g. SSSI selection guidelines) irrespective of whether or not it has yet been notified. Any regularly occurring population of a nationally important species which is threatened or rare in the region or county (local BAP). A regularly occurring, regionally or county significant population/number of any nationally important species..
Regional/high	Viable areas of key habitat identified in the Regional BAP or smaller areas of such habitat which are essential to maintain the viability of a larger whole. Viable areas of key habitat identified as being of Regional value in the appropriate Natural Area profile. Any regularly occurring, locally significant population of a species listed as being nationally scarce which occurs in 16-100 10km squares in the UK or relevant Natural Area on account of its regional rarity or localisation. A regularly occurring, locally significant number of a regionally important species. Sites which exceed the County-level designations but fall short of SSSI selection guidelines, where these occur.
County / Metropolitan/Medium	Semi-natural ancient woodland greater than 0.25 ha. County/Metropolitan sites and other sites which the designating authority has determined meet the published ecological selection criteria for designation, including Local Nature Reserves selected on County / metropolitan ecological criteria (County/Metropolitan sites will often have been identified in local plans). A viable area of habitat identified in County BAP. Any regularly occurring, locally significant population of a species which is listed in a County/Metropolitan “red data book” or BAP on account of its regional rarity or localisation. A regularly occurring, locally significant number of a County/Metropolitan important species.

Value/ sensitivity	Receptors
District / Borough	Semi-natural ancient woodland smaller than 0.25 ha. Areas of habitat identified in a sub-County (District/Borough) BAP or in the relevant Natural Area profile. District sites that meet the published ecological selection criteria for designation, including Local Nature Reserves selected on District/ Borough ecological criteria (District sites, where they exist, will often have been identified in local plans). Sites/features that are scarce within the District/Borough or which appreciably enrich the District/Borough habitat resource. A diverse and/ or ecologically valuable hedgerow network. A population of a species that is listed in a District/Borough BAP because of its rarity in the locality or in the relevant Natural Area profile because of its regional rarity or localisation. A regularly occurring, locally significant number of a District / Borough important species during a critical phase of its life cycle.
Local/ Low	Areas of habitat considered to appreciably enrich the habitat resource within the context of the Parish or neighbourhood (e.g. species-rich hedgerows). Local Nature Reserves selected on Parish ecological criteria.

Features with a value of local or above was considered to represent a 'Valued Ecological Receptor' (VER). Those features not meeting the criteria for VERs were classified as having either Site level or negligible ecological value.

Impact assessment

9.2.21 The impacts of the proposals have been predicted, taking into account different stages and activities within the development process. The significance of likely effects was determined by identifying those receptors likely to be affected. The features were evaluated to identify the important ones, i.e. those which, if their level of value reduced, national or local policies (or in some cases legislation) would be triggered. The nature of the individual and combined impacts were characterised on each important feature, to determine the longevity, reversibility and consequences for the feature in terms of ecological structure and function. Where it was concluded that an effect would be likely to reduce the value given to an important feature, it was described as significant. Therefore, the ecological significance of these impacts has then been assessed based upon the likely effect on the integrity or conservation status of each feature. The assessment of impact significance is undertaken both to identify the need for mitigation and also to assess residual impacts.

Significance Criteria

9.2.22 The ecological significance of these impacts has then been assessed, based upon the likely effect on the integrity or conservation status of the feature. The assessment of impact significance is completed both to identify the need for mitigation and also to assess residual impacts.

9.2.23 The significance of likely effects was determined by:

- identifying those ecological features likely to be affected;
- evaluating them to identify the important ones (i.e. those which, if their level of value reduced, national or local policies (or in some cases legislation) would be triggered); and
- characterising the nature of the individual and combined impacts on each important feature, to determine longevity, reversibility and consequences for the feature in terms of ecological structure and function.

9.2.24 Where it was concluded that an effect would be likely to reduce the value given to an important feature, it was described as significant.

9.2.25 To provide some consistency with other chapters, the significance of impacts is also assessed using a series of matrices. The approach is a three-stage process where the value of the ecological receptor and the magnitude of the impact are cross-tabulated to identify impact significance. The details are set out in Tables 9.3 (sensitivity), above and 9.4 (magnitude) and 9.5 (significance) below.

Table 9.4 Magnitude of Impact

Magnitude	Description
Large	The proposal (either on its own or with other proposals) may adversely affect the integrity of the site/habitat, in terms of the coherence of its ecological structure and function, across its whole area, that enables it to sustain the habitat, complex of habitats and / or the population levels of species of interest.
Moderate	The sites/habitats integrity will not be adversely affected, but the effect on the site/habitat is likely to be significant in terms of its ecological objectives. If, in the light of full information, it cannot be

	clearly demonstrated that the proposal will not have an adverse effect on integrity, then the impact should be assessed as major negative.
Small	Neither of the above apply, but some minor negative impact is evident.
Negligible	No observable impact in either direction.
Positive	Impacts which provide a net gain for wildlife overall.

Table 9.5: Significance of ecological impacts

MAGNITUDE	**SENSITIVITY**		
	High	**Medium**	**Low**
Large	Major	Major	Moderate
Moderate	Major	Moderate	Minor
Small	Moderate	Minor	Minor
Negligible	Minor	Negligible	Negligible

9.3 RELEVANT POLICY & LEGISLATION

National Planning Policy Framework (March 2012)

9.3.1 Guidance on nature conservation planning policy is provided in the 'Conserving and enhancing the natural environment' section of the National Planning Policy Framework (NPPF).

NPPF Core Planning Principles

9.3.2 Paragraph 17 of the NPPF states that "within the overarching roles that the planning system ought to play, a set of core land–use planning principles should underpin both plan–making and decision taking". Of the 12 principles outlined, the following are relevant to ecology and nature conservation:

- contribute to conserving and enhancing the natural environment and reducing pollution. Allocations of land for development should prefer land of lesser environmental value, where consistent with other policies in this framework;

- promote mixed developments and encourage multiple benefits from the use of land in urban and rural areas, recognising that some open land can perform many functions (such as for wildlife, recreation, flood risk mitigation, carbon storage or food protection).

Conserving and Enhancing the Natural Environment

9.3.3 At Section 11, paragraph 109 states "the planning system should contribute to and enhance the natural and local environment by:

- protecting and enhancing valued landscapes, geological conservation interests and soils;
- recognising the wider benefits of ecosystem services;
- minimising impacts on biodiversity and providing net gains in biodiversity where possible, contributing to the Government's commitment to halt the overall decline in biodiversity, including by establishing coherent ecological networks that are more resilient to current and future pressures;

- preventing both new and existing development from contributing to or being put at inacceptable risk from or being adversely affected by unacceptable levels of soil, air, water or noise pollution or total instability; and
- remediating and mitigating despoiled, degraded, derelict, contaminated and unstable land where appropriate.

9.3.4 Paragraph 113 states "Local Planning Authorities should set criteria based policies against which proposals for any development on or affecting protected wildlife or geodiversity sites or landscape areas will be judged. Distinctions should be made between the hierarchy of international, national and locally designated sites so that protection is commensurate with their status and gives appropriate weight to their importance and the contribution that they make to wider ecological networks".

9.3.5 Paragraph 118 addresses the conservation and enhancement of biodiversity and requires LPAs to "conserve and enhance biodiversity by applying the following principles:

- if significant harm resulting from development cannot be avoided (through locating on an alternative site with less harmful impacts), adequately mitigated, or as a last resort, compensated for, then planning permission should be refused;
- proposed development on land within or outside a Site of Special Scientific Interest likely to have an adverse effect on a Site of Special Scientific Interest (either individually or in combination with other developments) should not normally be permitted. Where an adverse effect on the site's notified special interest features is likely, an exception should only be made where the benefits of the development, at this Site, clearly outweigh both impacts that it is likely to have on the features of the Site that make it of Special Scientific Interest and only broader impacts on the National Network of Sites of Special Scientific Interest;
- development proposals where the primary objective is to conserve or enhance biodiversity should be permitted;
- opportunities to incorporate biodiversity in and around developments should be encouraged;
- planning permission should be refused for development resulting in the loss or deterioration of irreplaceable habitats, including ancient woodland and the

loss of aged or veteran areas found outside ancient woodland unless the need for, and benefits of, the development in that location clearly outweigh the loss; and

- The following wildlife sites should be given the same protection as European Sites:

  -potential Special Protection Areas and possible Special Areas of Conservation;

  -listed or proposed Ramsar Sites; and

  - sites identified, or required, as compensatory measures for adverse effects on European Sites, Potential Special Protection Areas, possible Special Areas of Conservation, and listed or proposed Ramsar Sites.

Swindon Borough Local Plan 2011 (2006)

9.3.6 Saved policies in the Swindon Borough Local Plan 2011 that are of relevance include:

Biodiversity: Local Sites Policy ENV17

Development within, adjoining or directly affecting Local Nature Reserves and Sites of Importance for Nature Conservation (also known as County Wildlife Sites) or species and habitats noted in Local Biodiversity Plans (Wiltshire BAP 2002 and Swindon BAP 2005), which is likely to have a detrimental impact, either individually or in combination with other proposals, shall only be permitted where it can be demonstrated that the reasons for development outweigh the need to safeguard the substantive nature conservation value of the site. Where development is permitted the Local Planning Authority shall seek:

a) to maximise the protection and enhancement of the site's intrinsic features of nature conservation value; as well as

b) compensatory measures to replace the loss of existing wildlife and habitats on at least a pro rata basis.

Policy ENV18 Habitat and Species Protection

Development shall only be permitted where it can be demonstrated that features of intrinsic nature conservation value, including species specially protected by law, are either protected or enhanced, or if this cannot be achieved compensatory replacement measures are provided on at least a pro rata basis, and secured by conditions or planning obligations.

Policy ENV24 Riparian and Other Buffer Zones

Development that includes or adjoins rivers, watercourses and other water bodies shall only be permitted where development free buffers extending outwards from the feature are included in the proposal, their scale and width reflecting the intrinsic natural value of each specific zone.

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9.3.7 Relevant policies include:

Policy EN1: Green Infrastructure Network

a. Development shall protect and enhance green infrastructure and assets as identified in Appendix 5. This includes the requirement that development must provide for the protection and integration of visually or ecologically important existing trees, hedges and woodlands. Development that would result in the loss or deterioration of irreplaceable habitats, including ancient woodland and the loss of aged or veteran trees found outside ancient woodland will only be permitted where the need for, and benefits of, the development in that location clearly outweigh the loss.

b. Development shall design green infrastructure and integrate with existing green corridors identified on the Proposals Map, to maximise its connections and functions and ensure the sustainable maintenance and management of it.

Policy EN2: Community Forest

Development shall contribute towards the aims and objectives of the Great Western Community Forest (GWCF) in Swindon. This will be achieved by:

- ensuring a net increase in tree cover through the planting of new woodland and trees;
- creating or enhancing habitats for biodiversity, including built structures in accordance with Polices EN1 and EN4; and
- ensuring access to local woodlands and opportunities for communities and businesses to benefit from GWCF.

Policy EN4: Biodiversity and Geodiversity

a. Development will avoid direct and indirect negative impacts upon biodiversity and geodiversity sites as identified on the Proposals Map. This will be achieved through sensitive site location and layout, and by maintaining sufficient buffers and ecological connectivity with the wider environment. Damage or disturbance to local sites will generally be unacceptable, other than in exceptional circumstances where it has been demonstrated that such impacts are:

- unavoidable and reduced as far as possible or are outweighed by other planning considerations in the public interest, and
- where appropriate compensation measures can be secured.

b. All development, where appropriate shall protect and enhance biodiversity and provide net local biodiversity gain. Where this is demonstrably not achievable, mitigation and compensation measures shall be considered and agreed.

Legislative Framework

9.3.8 The following legislation and European Directives afford protection to wildlife and have been used in particular to inform this assessment:

- Natural Environment and Rural Communities Act 2006 (NERC);
- Wildlife and Countryside Act (1981) (as amended) (WCA);
- The EC Birds Directive (Directive 79/409/EEC), as translated into UK law by The Habitat and Species Regulations 2010 (as amended);
- The EC Habitats Directive (Directive 92/43/EEC) as translated into UK law by The Habitat and Species Regulations 2010 (as amended);
- The Protection of Badgers Act (1992).

9.3.9 Section 41 of the Natural Environment and Rural Communities Act 2006 places a duty on the Secretary of State to publish, review and revise lists of living organisms

and types of habitat in England that are of principal importance for the purpose of conserving English biodiversity, and to consult Natural England before doing so.

9.3.10 The S41 list is used to guide decision-makers such as public bodies, including local and regional authorities, in implementing their duty under section 40 of the Natural Environment and Rural Communities Act 2006, to have regard to the conservation of biodiversity in England, when carrying out their normal functions.

9.4 BASELINE CONDITIONS

Statutory Designations

9.4.1 No internationally designated sites occur within 10km of the Site boundary. The closest site, North Meadow and Clattinger Farm SAC, occurs some 11.3km to the north-east.

9.4.2 The SAC is considered to be one of the best areas in the UK for unimproved lowland hay meadow (*Alopecurus pratensis* and *Sanguisorba officinalis*), with a rich variety of grasses and hay meadow flowers. North Meadow supports Britain's largest population of snake's head fritillary (*Fritillaria meleagris*), which is now rare throughout its range, representing >90% of the UK population. The plant species are important in supporting varied insects and birdlife, including waders in winter. The species-rich unimproved hay meadows are on moderately fertile soils of river and tributary floodplains. Seasonal flooding maintains an input of nutrients and is vital for the integrity of the site and the variety of flora present.

9.4.3 As a consequence of the site's internationally important vegetation communities, the site is considered to be of International importance

9.4.4 Only a single further statutory designated site, Stanton Park LNR, located some 1.9 km to the north of the Site, occurs within the Study Area.

Non-Statutory Designations

9.4.5 The current system used in Wiltshire for non-statutory designated sites refers to County Wildlife Sites (CWS), several of which occur close to the Site and are of County/Medium significance. The Wiltshire and Swindon Biological Records Centre (WSBRC) highlighted the presence of a total of three County Wildlife Sites (CWSs) within 1km of the Site. A summary of each is provided in Table 9.6 below.

Table 9.6: County Wildlife Site Summary

Site Name	Location	Status	Source
River Cole	SU217880 – flows through and adjacent to Site	County Wildlife Site (CWS)	Wilts & Swindon Biological Records Centre (WSBRC)

Site Name	Location	Status	Source
River Cole Meadow	SU188859 – c.1km south to west of A419	CWS	WSBRC
Pits north of South Marston	SU195887 – 700m north of Site	CWS	WSBRC

9.4.6 Notable species records, such as those that receive some degree of statutory protection or non-statutory policy protection, were provided by a number of organisations including the BSBI, English Nature (now Natural England) and WSBRC. A full list is provided in **Appendix 9.1 B** and a summary is provided in Table 9.7 below and Figure 1: Phase 1 Habitat Plan (**Appendix 9.1**)

Table 9.7: Notable Species Records

Species	Location	Status	Source
Black poplar *Populus nigra*	2km to the south on River Cole and Liden Brook	Vice-county Rare (VC Rare)	BSBI
Yew *Taxus baccata*	SU1988 – 600m north of boundary	Wilts Rare Plant List	WSBRC
Columbine *Aquilegia vulgaris*	SU195883 – 300m north of Site	Not given	WSBRC
Mouse-ear hawkweed *Pilosella agg*	SU192885 – 700m north of Site	Not given	WSBRC
Prickly lettuce *Lactuca serriola*	SU1888 – 1km northwest of Site	Not given	WSBRC
Dwarf spurge *Euphorbia exigua*	SU1987 – within Site	IUCN Near Threatened	BSBI
Pale toadflax *Linaria repens*	SU197871 – within Site	VC Scarce	BSBI
Butcher's broom *Ruscus aculeatus*	SU2083 – 200m east of boundary	Wilts Rare Plant List	WSBRC
Great burnet *Sanguisorba officinalis*	To the south and east of Site	Wilts Rare Plant List	WSBRC
Goldenrod *Solidago virgaurea*	SU217854 – 2km east of Site	Wilts Rare Plant List	WSBRC
Barn owl *Tyto alba*	SU189855 – 500m west of Site	Schedule 1 Wildlife and Countryside Act 1981	WSBRC
Water vole *arvicola amphibus*	Various locations in and around Site and wider area.	Schedule 5 Wildlife and Countryside Act 1981 (W&CA-5)	WSBRC
Bat sp. *Chiroptera sp.*	SU193838 – 100m west of Site	W&CA-5	WSBRC
Serotine *Eptesicus serotinus*	Two locations both over 300m from Site	W&CA-5	WSBRC
Common pipistrelle *Pipistrellus pipistrellus*	SU1885 – 2.5km west of Site	W&CA-5	WSBRC

Species	Location	Status	Source
Pipistrelle sp.	Various locations in and around Site and wider area.	W&CA-5	WSBRC
Brown long-eared *Plecotus auritus*	SU192877 – 200m north of Site	W&CA-5	WSBRC
Brown hare *Lepus europaeus*	A number of locations including within Site boundary	UKBAP species	WSBRC
Otter *Lutra lutra*	Various locations on River Cole and Liden Brook.	W&CA-5	WSBRC
Badger *Meles meles*	Various locations including Site and wider area.	Badgers Act.	WSBRC
Water shrew *Neomys fodiens*	SU188889 – 300m west of Site	Proposed County Notable	WSBRC
Slow worm *Anguis fragilis*	SU1885 – 1km west of Site	W&CA-5	WSBRC
Viviparous lizard *Lacerta vivipara*	Two locations: 1km west and 700m east of Site	W&CA-5	WSBRC
Grass snake *Natrix natrix*	Two locations: to north and west of Site	W&CA-5	WSBRC
White-clawed crayfish *Austropotamobius pallipes*	SU2187 – within and 100m north of Site	W&CA-5	WSBRC
Brown argus *Aricia agestis*	SU228826 – The Coombs, Hinton Parva SSSI. 4.8km south of Site.	Not listed	English Nature
Chalk-hill blue *Lysandra coridon*	SU228826 – The Coombs, Hinton Parva SSSI. 4.8km south of Site.	W&CA-5	English Nature
Marbled white *Melanargia galathea*	SU228826 – The Coombs, Hinton Parva SSSI. 4.8km south of Site.	Not listed	English Nature
Land winkle sp. *Pomatias elegans*	SU228826 – The Coombs, Hinton Parva SSSI. 4.8km south of Site.	Not listed	English Nature
Snail sp. *Helicella itala*	SU228826 – The Coombs, Hinton Parva SSSI. 4.8km south of Site.	Not listed	English Nature

Evaluation of Valued Ecological Receptors

Overview

9.4.7 Broad habitat types within the Site are identified on Figure 1: Phase 1 Habitat Plan (**Appendix 9.1**) and further habitat descriptions and species lists are provided in **Appendix 9.1 A and B,** respectively.

9.4.8 A range of habitats are present within the Site and are typically heavily influenced by the current agricultural management of the Site. The Site is dominated by arable fields and improved pasture divided by hedgerows. There are occasional species-poor, semi-improved grasslands present in association with field boundaries and areas of less intensive management. Mature trees are frequently associated with hedgerows.

9.4.9 Significant parts of the Site are planted with willow coppice, which, it is understood, was planted as a source of biofuel, although remains un-harvested.

9.4.10 The South Marston Brook, a tributary of the River Cole, forms a significant feature within the Site where it flows from north to east before its confluence with the River Cole some 500m to the east.

Habitats

Broad-leaved Plantation Woodland

9.4.11 Broad-leaved plantation woodland occurs scattered within the Site in the form of recently planted, willow dominated woodland. It is understood that once cropped this is intended as a sustainable biofuel resource. In its eastern extent, adjacent to the eastern Site boundary the woodland is of a young age (c. 5-8yrs) and is associated with a ruderal and coarse grassland margin and ground flora.

9.4.12 A further area of est. 10-15yrs growth also occurs to the south of Rowborough Farm where evenly aged drawn willow specimens including crack *Salix fragilis* and osier willows *S. viminalis* with occasional wych elm *Ulmus glabra* provided a high level of shade, with consequently little real understory. The ground flora was of low diversity and generally dominated by nettle *Urtica dioica* with other occasional tall ruderal herbs.

9.4.13 The woodland units are relatively homogenous and lack structural or species diversity and are therefore limited in interest. Nevertheless they may provide a relatively undisturbed habitat for wildlife for shelter and forage; and facilitate dispersal though the landscape for more secretive species. This habitat type is therefore considered to be of Local/Low nature conservation significance.

Coniferous Plantation Woodland

9.4.14 A small Christmas tree plantation dominated by Norway Spruce *Picea Abies* is located to the northeast of Rowborough Farm. The trees are densely planted and relatively tall to 5-10m in height allowing little associated vegetation to persist and as a result the woodland is considered to be of negligible nature conservation value.

Semi-improved Grassland

9.4.15 Semi-improved grassland is generally highly localised within the Site and persists where intensive agricultural management is reduced. All are species–poor in nature and where ungrazed, such as adjacent to Nightingale Lane, are dominated by a range of common coarse grasses such as cock's-foot *Dactylis glomerata*, false oat-grass *Arrhenatherum elatius*, common couch *Elitrigia repens* with frequent tall ruderal herbs such as cow parsley *Anthriscus sylvestris* and hogweed *Heracleum sphondylium*. Occasional species included red clover *Trifolium pratense* and meadow barley *Hordeum secalinum.*

9.4.16 Where grazed, such as the small paddocks adjacent to Rowborough Farm and Manor Farm, constant species included perennial ryegrass *Lolium perenne*, Yorkshire fog *Holcus lanatus* and crested dog's-tail *Cynosurus cristatus* with a range of grazing tolerant herbs such as creeping buttercup *Ranunculus repens*, white clover *Trifolium repens*, cut-leaved crane's-bill *Geranium dissectum*, common cat's-ear *Hypochoeris radicata* and greater plantain *Plantago major*.

9.4.17 Some small field compartments to the south of South Marston also support a semi-improved grassland sward. The first is located just to the north of Priory Farm where a small number of pear trees exist. Although some species indicative of less intensive management were noted such as meadow vetchling *Lathyrus pratense* and common sorrel *Rumex acetosa*, coarse grasses were overwhelmingly dominant with

false oat grass and common couch of high frequency. The diversity and frequency of herbs was low.

9.4.18 Species poor grassland habitats are generally ubiquitous in the wider countryside and where of limited scale, such as those within the Site and identified above, are not of significant nature conservation value.

Marshy grassland

9.4.19 Small patches of marshy grassland occur to the south of South Marston, where damp depressions/ditches occur. A suite of grassland species typical of water-logged conditions were prevalent and included frequent abundant creeping bent *Agrostis stolonifera*, silverweed *Potentilla anserina*, tufted hair grass *Deschampsia cespitosa*, floating sweet grass *Glyceria fluitans*, marsh foxtail *Alopecurus pratensis* and hairy sedge *Carex hirta*.

9.4.20 Marshy grassland areas whilst not of high value do appreciably enrich biodiversity where they occur and provide a local increase in micro habitat diversity at the field compartment scale. As a result this habitat type is considered to be of Local/Low significance.

Tall Ruderal Vegetation

9.4.21 Tall ruderal vegetation is located frequently along hedgerows bases as is typical where arable and improved grassland predominate. The hedge bottom flora is largely ruderal in its composition with bramble *Rubus fruticosus* agg., common nettle *Urtica dioica* and cleavers *Galium aparine* all frequent and often abundant throughout the hedge network. Cock's-foot, Yorkshire-fog *Holcus lanatus* and false oat-grass were all key grass species.

Water Bodies – Running Water

9.4.22 Two water courses, the River Cole and a tributary of it - the South Marston Brook - are associated with the Study Area. The River Cole exists to the east of the Application Site. Its banks are typically steep to around 2m height and supported a marginal flora including occasional marsh woundwort *Stachys palustris*, reed canary grass, gypsywort, water mint *Mentha aquatica* and amphibious bistort *Polygonum amphibium*. In-channel vegetation included occasional stands of branched bur-reed

Sparganium erectorum, fools watercress *Apium nodiflorum* and yellow flag iris *Iris psuedacorus*.

9.4.23 The South Marston Brook flows through the Site from South Marston in the north through to its confluence with the River Cole in the east. It is heavily shaded throughout much of its length by adjacent ruderal vegetation, trees and scrub. The banks range from steep to vertical and are undercut in places.

9.4.24 Tall herb vegetation including abundant nettle, greater willowherb and coarse grasses typically dominated the bankside where more open in character. A marginal flora was present including common comfrey, greater willowherb and meadow sweet. An associated ditch supported a more diverse wetland flora including watercress *Rorippa nasturtiun-aquaticum*, brooklime *Veronica beccabunga*, reedmace *Typha latifolia*, greater pond sedge *Carex riparia*, fools water cress and common reed *Phragmites australis*.

9.4.25 Both the River Cole and South Marston Brook support a range of natural morphological features and as a result of the diversity of physical characteristics such as depth, temperature, flow-rate and substrate, are likely to support a diverse assemblage of species. They also provide a relatively continuous corridor of semi-natural habitat managed at fairly low intensity and as a result are likely to be used by both aquatic and non-aquatic species. Both are designated as the River Cole CWS and are considered to be of County/Medium significance.

Water Bodies – Running Water

9.4.26 A number of ponds were identified within or just outside of the Study Area: both to the north of the A420 and to the south. The habitats present vary from deep ponds with substantial bankside vegetation to ephemeral shallow ponds or recent man-made ponds. A summary of pond characteristics is provided in Table 9.8 below:

Table 9.8: Pond Characteristics

Pond reference	Notes
P1	Deep pond with shallow gradient, 10 x 16 m. Heavily shaded by willow and scrub. Aquatic vegetation included *Callitriche stagnalis, Vernica beccabunga* and *Lemna minor*. Fallen branches and litter were abundant
P2	Large pond of 20 x 15 m with lush dense emergent vegetation

Pond reference	Notes
	included *Rorippa nasturtiun-aquaticum*, water crowfoot, *Glyceria maxima* and *Iris psuedacorus* with *L. minor*.
P3	Recently (part?) drained ponds with bare exposed draw-down zone evident. Some *Juncus inflexus* at former water level and part submerged tree stumps.
P4	Heavily shaded pond with many overmature willows. Vegetation includes *Apium nodiflorum, Ranunculus sp., R. flammula*, and *Lemna* sp. Adjacent wet ditch supporting marshy grassland with abundant *Potentilla anserina*, *D. cespitosa, Carex hirta* and *Agrostis stolonifera*. Dry in summer.
P5	Large ornamental pond 17 x 5 m not visited in 2006. Overshaded by mature trees with little aquatic flora.
P6	Over shaded pond, 15 x 6 m. Overshaded on three sides by grown out hedgerows and ash trees. Aquatic vegetation limited to *Iris psuedacorus, Lemna and Carex obtrubae*.

9.4.27 Despite many of the ponds being over-shaded and/or being only of seasonal nature due to succession/silt, ponds at all successional stages can be important to wildlife including notable and/or protected species such as great crested newts and other amphibians. As a result the network of ponds within the Site is considered to be of Local/Low significance.

Mature Trees

9.4.28 Standard trees were mainly associated with hedgerows and the South Marston Brook, ash were generally the dominant species along hedgerows, although pollarded crack willows were also frequent particularly along the course of the Brook. Overall, the tree stock was in good condition and well distributed throughout the Site; enhancing the structural diversity of the Site and providing a habitat for a range of species excluded from the more intensively managed agricultural habitats. Other trees of less frequent occurrence included occasional field maple, small-leaved lime and Horse chestnut.

9.4.29 Mature trees within the Site were considered to be of Local/Low significance.

9.4.30 Over-mature and near veteran specimens were present in the form of a number of crack willows along the brook where signs of historic or even recent management were noted. As is typical and fairly common for the species, many had large hollowed out main trunks showing signs of different forms of rot, the fruiting bodies of

fungus, the presence of holes used by birds and small mammals, and evidence of use by saproxylic invertebrates.

9.4.31 More notable trees included a single veteran ash pollard (refer to T160, Arboricultural Assessment (FPCR 2013)) of large girth and features indicative of veteran status. Veteran trees within the Site are considered to be of District/Medium significance.

Hedgerows

9.4.32 Hedgerows within the Site were of variable quality, largely due to the variation in management practice currently in operation. During surveys undertaken in 2002 and 1999, a total of nine hedgerows qualified as important under the Wildlife and Landscape criteria of the Hedgerow Regulations 1997 and, as a result of the review undertaken in 2006, which did not identify any significant changes, their status is likely to be maintained. Hedgerows identified as important include H49, H50, H54, H57, H58, H71, H74, H76 and H83 and are identified within Figure 3 (**Appendix 9.8**).

9.4.33 Typically hedgerows were generally dominated by hawthorn with abundant blackthorn *Prunus spinosa*, elm *Ulmus* spp, and dog rose *Rosa cannina*. .

9.4.34 A summary of hedgerow features and nature conservation interest is provided in Table 9.9 below:

Table 9.9: Hedgerow Summary

Hedgerow No	Species	Dimensions	Hedgerow Regs
H1	Cm, Ps, Fe, Ulmus sp	2 x1	
H2	Ac, Cm, Ulmus sp, Fe, Sn	1.5 x 1	
H3	Cm, Sn, Ulmus sp, Fe, Ap	5 x 4 (max. 7)	
H4	Cm, Ulmus sp	5 x 4	
H5	Cm, Sn, Ulmus sp	1.5 x 2	
H6	Cm	3 x 3	
H7	Cm, Mainly bramble	1 x 2	
H8	Ps, Cm, Ulmus sp, Sn,	5 x 6	
H9	Cm, Ulmus sp, Ms	5 x 6	
H10	Cm, Ulmus sp, Ms, Sn,	6 x 6	
H11	Cm, Ulmus sp, Ms	5-7 x 4	
H12	Ca, Cm, Ulmus sp, Fs	5 x 4	
H13	Qr, Cm, Ulmus sp, Fe	7 x 6	
H14	Sn,Ulmus sp, Cm, Ps, Fe, Sf	4-8 x 4	
H15	Predominantly Ulmus sp up to the Western end and Eastern end Ps, Cm, Ac, Sn.	8 x 6	
H16	Cm, Some Ms Ulmus sp	5 x 4	

Hedgerow No	Species	Dimensions	Hedgerow Regs
H17	Cm, Ulmus sp/ Ms, Some substantial Ac	5-7 x 6	
H18	Cm, Ms, Ulmus sp, Small amount of Sf	4-6 x 4-6	
H19	Ulmus sp. Some more Sf, Cm	6 x 8	
H20	Cm, Ac, Fe	4-5 x 4	
H21	Cm, Ia, Ap, Pa	max. 8 (trees) x 6	
H22	Cm	4 x 4	
H23	Cm, Ulmus sp, Fe, Ms	6 x 6	
H24	Some Cypress and some Cm	0.5 -3 x 1	
H25	Sc, Cm, Ac	3 x 2	
H26	Cm, self set Fe, Sn, Ulmus sp	4-7 x 4-6	
H27	Cm, Ps, Ulmus sp	5x4	
H28	Fe, Cm, Salix sp	8x4	
H29	Cm, Ulmus sp	2x2	
H30	Cm, Ulmus sp	2x2	
H31	Cm, Ps	2x2	
H32	Cm, Ps	1.5x2	
H33	Cm, Ps	2x2	
H34	Cm, Ps, Ulmus sp	4-8x6	
H35	Cm, Ps	2x2	
H36	Cm, Ps	2x2	
H37	Cm, Ps	2x2	
H38	Ps	4 x 3	
H39	Ulmus sp	Up to 7 x 4-6	
H40	Cm	1.5x2	
H41	Ac, Ps, Cm	Up to 5x	
H42	Cm, Sn, Ulmus sp, Sc	Up tp 7x4-6	
H43	Ac, Ps, Cm, Ulmus sp	Up to 7x6-8	
H44	Cm, Ulmus sp, Sn	3-4x3	
H45	Cm, Ps	2x2	
H46	Cm, Ac	Up to 8 x 8	
H47	Cm, Some coppice Ca	4 x 4-6	
H48	Cm, Ps, Ulmus sp	3 x 2	
H49	Cm, Some Ps, Ulmus sp	2x2	yes
H50	Cm, Ps, Ulmus sp, Fe	Max 2x2	yes
H51	Predominantly Cm	6x6	
H52	Cm, Ps, Ac, Ms	5-7x4	
H53	Ac, Ps, Cm	Max 5x4	
H54	Cm, Ac, Ulmus sp	2x2	yes
H55	A lot of Sn, Cm, Ps,Several coppice Ca at West end	4x2	
H56	Almost entirely Sn and Ulmus sp,.Occasional Cm, Ps	2-5x2	
H57	Cm, Ps, Ulmus sp, Ca, Sn,	Max 6x4	yes
H58	Coppice Ca, Sn, Sc, Damson	5x4	yes
H59	Cm, Sn	1.5x3	
H60	Ac, Ps, Cm, Lv	2x4	
H61	Ca, Cm, Ps, Ac	2x4	
H62	Sn, Cm, Ps, Ac, Fe	3x3	

Hedgerow No	Species	Dimensions	Hedgerow Regs
H63	Cm, Ps, Ulmus sp, Sn	2x4	
H64	Sn, Cm	1x2	
H65	Ps, Cm, Ac,Ms	5x3/4	
H66	Cm, Ac, Ps, Ulmus sp	2x2	
H67	Cm, Ps	2x2	
H68	Ulmus sp,Cm, Ac, Sn	4x1	
H69	Ulmus sp, Cm, Ac, Sn	Up to 8x6	
H70	Ps, Cm, Ac. Some standard Fe up to 7m	4x2	
H71	Sn, Cm, Ac	4x1	yes
H72	Sn, Cm	2.5x2	
H73	Ac, Ulmus sp, Cm, Ps	2 (with growth)x2	
H74	Ps, Sn, Cm, Ac, Ms, Ca	3x1	yes
H75	Ps, Cm, Ac, Fe, Ca	2x2	
H76	Ulmus sp, Sn, Cm	2-3(5)x 4	yes
H77	Ulmus sp, Cm, Ps	3 x4	
H78	Ps, Ac, Ulmus sp, Cm	4 x3	
H79	Ms, Ps, Sc	Up to 5 x 20	
H80	Mostly Sn, Cm	34x4	
H81	Sn, Cm, Fe	5-6 (8) x 6	
H82	Predominantly Cm, Ps, Sn, coppice Ca, Ms	5x6	
H83	Ps, Cm	4x4	yes
H84	Ag, Ps, Cm, Ms, Sn	4x4	
H85	Ac,Ulmus sp, Ps, Cm	Up to 5x2	
H86	Cm	5x6	
H87	Cm, Sn, Ulmus sp, Ps	5x4	

Key: Cm *Crateagus monogyna*, Ps *Prunus spinosa*, Ca *Corylus avellana*, Ms *Malus sylvstris*, Ac *Acer campestre*, Sn *Sambucus nigra*, Sc *Salix caprea*, Fe *Fraxinus excelsior*, Ag *Alnus glutinosa*, Lv *Ligustrun vulgare*, Pa *Prunus avium*, Ap Acer *psuedoplatanus*, Qr *Quercus robur*, Sf *Salix fragliis*

9.4.35 All hedgerows are considered to be of at least Local/Low significance as, to varying degrees they enrich biodiversity at a local scale and, where composed predominantly of native species, qualify as priority habitats as listed under the NERC Act.

Fauna

9.4.36 In additional to desk study information, which revealed the presence of a number of protected and/or otherwise notable species, a range of further faunal surveys have been conducted within the Site since 2002. These are provided in detail in **Appendices 9.2 to 9.7**. A summary of identified potential receptors is outlined below.

Amphibians

9.4.37 There are no statutory or non-statutory sites within 1km of the Site boundary that have been designated on the basis of the amphibian populations they support.

9.4.38 Surveys completed within the Site confirmed the presence of great crested newt in two ponds (P1 and P4) within the Site. A summary of results is provided in Table 9.10 below:

Table 9.10: GCN Results Summary

Pond reference	HSI score	Notes	GCN presence?
P1	0.68 (Average)	GCN confirmed in 2002 (peak count 4) and 2006 (peak count 1). Seasonal/ occasional drying. None in 2012.	Yes
P2	0.91 (Excellent)	Small population of smooth newt present	No
P3	0.73 (Good)	Fish (stickleback) present	No
P4	0.71(Good)	GCN confirmed in 2006 (peak count 9), and 2012 (peak count 1). Seasonal/ occasional drying.	Yes
P5	Unknown	Surveyed 2002 only; no access subsequently	No
P6	0.71(Good)	Small population of smooth newt found	No

9.4.39 The population size class assessment as defined within the English Nature guidance (August 2001) is as follows:

- Small – for maximum counts up to 10
- Medium – for maximum counts between 11 – 100
- Large – for maximum counts over 100

9.4.40 The above classification is a broad category used to indicate the comparative status and importance of a population.

9.4.41 Ponds P1 and P4 were observed to support no more than a small population of GCNs, as no more than nine GCNs were found on any survey occasion. Furthermore the populations in both ponds appear to be declining with only a single newt, in Pond 4, recorded in 2012. The distance between ponds in addition to the partial barrier

formed by the South Marston Brook is likely to indicate the two populations are distinct, with no immigration and emigration between ponds.

9.4.42 Adjacent commercial turf production and improved grassland areas, which form the majority of habitat surrounding the ponds, are considered to be of low to moderate value depending on the management regime in operation. However, the more established hedgerows and coarse grassland and immediately adjacent habitats such as the plantation woodland around P1 are considered to be of moderate to high value especially given their potential value for shelter and foraging.

9.4.43 Previous surveys of wider area suggest that great crested newts (GCN) are fairly widespread but localised to the east and south of Swindon. The populations centred within the Site are also small and, in this context, are considered to be of no more than District/Medium significance.

9.4.44 Further detailed survey information is provided in **Appendix 9.2**.

Bats

9.4.45 There are a number of bat records in the vicinity of the Application Site including one record of a pipistrelle species *Pipistrellus* sp to the south of the Site, to the northwest of Lotmead Farm. Two further records of pipistrelle species exist approximately 300m to the north of the Site boundary within South Marston.

9.4.46 During activity surveys completed within the Site during 2006, three species of bat were recorded to the north of the A420: Noctule *Nyctalis noctua*, common pipistrelle *Pipistrellus pipistrellus* and Brown Long-eared *Plecotus auritus*. Areas of particular importance were considered to be associated with the east of the Site to the south of Nightingale Wood where hedgerows (H71, H74, H76, H73, H77 & H82) appear to act as corridors of movement between the river corridor and Nightingale Wood. Further areas of higher activity were noted immediately south of South Marston and in association with the roadside hedgerows that bisected the Site in this area (TN 9).

9.4.47 During the transects results recorded over 2013, the total number of bat contacts ranged from a maximum of 26 to a minimum of 4 bats; a relatively low total given the scale and intensity of survey effort.

9.4.48 A total of 7 bat species/species groups were recorded including; common pipistrelle, noctule, brown long-eared, a Myotis species, Soprano pipistrelle, a pipistrelle species and another unknown species. The most recorded species was common pipistrelle by a large margin. All of the recorded bat species were relatively common and abundant throughout the region.

9.4.49 Despite the routes being similar on each occasion, the areas of peak activity varied from survey to survey and were identified as being:

- H24 to north.
- H50 and associated house to east.
- Southern area of H53/watercourse
- H83 on eastern boundary.
- Southern stretch of H82
- H8 to south,
- H17 to south.
- H43 in south centre of the site,
- River Cole to east (off site),
- H82 to east,
- H60 to north.
- H9/H17 in centre of route,
- H16 to north.
- South Yew Cottages (road through centre of area,
- H9/H17 to south,
- H33in the east of route
- H85 on eastern boundary,
- H44 /H45 west of Rowsborough farm.

9.4.50 Static bats detector surveys identified a similar suite of species, with the addition of four species recorded in very low numbers: Serotine *Eptesicus serotinus* (9 passes over two survey occasions), Natterers *Mtyotis natterii* (single pass), Barbastelle *Barbastella barbastellus* (7 passes over four survey occasions), and Nathusius pipistrelle *Pipistrellus nathusiis* (single pass).

9.4.51 Despite only a relatively moderate level of bat activity being recorded during the activity surveys the site was identified during both transect and static bat detector surveys as supporting a relatively rich assemblage of bats/bats species groups

including some relatively rare species (Nathusius pipistrelle and Barbastelle). These were however only recorded rarely suggesting that the site is not of importance to any wider population of either species. Nevertheless the site does support occasional roosts for at least two species and some areas of foraging such as along the river and brook corridors. As a result the site is considered to be of district level significance for bats.

9.4.52 A range of features, including mature trees and buildings, were inspected to identify features of potential use by roosting bats. Where these were identified as being significantly directly or indirectly affected by proposals further surveys were completed. These included a total of 15 mature trees subject to aerial inspection and a stable block associated with the Rowborough Farm complex.

9.4.53 Only one tree, T50 was confirmed as having a roost site after the aerial assessment in July 2013. The tree had some woodpecker holes present with one hole having 15-20 x suspected Noctule droppings within. The droppings appeared to be fairly fresh indicating the roost was used recently. However, no bats were present at the time suggesting that any further nocturnal surveys would be unlikely to reveal further information and that the roost is only an occasionally used transitory roost. DNA analysis of droppings was inconclusive with too little bat DNA present to confirm species.

9.4.54 Overall the trees within the survey area are considered to represent Local value for bat populations.

9.4.55 Similarly, nocturnal surveys of the Rowborough Farm stable block also confirmed the presence of a bat roost, with single common pipistrelle bats seen to emerge on two of three survey occasions (with one occasion comprising both dusk emergence and dawn re-entry).

9.4.56 The roost comprised a small number of emerging common pipistrelle bats with a peak count of only 1. Given the lack of internal evidence this roost status is likely to be a small occasionally used roost site, occupied by small numbers (less than 5) of adult bats. Overall the buildings within the survey area are considered to represent Local value for bats.

9.4.57 Further detailed bat survey information is provided in **Appendix 9.3**

Reptiles

9.4.58 Existing records of grass snake, slow worm and common lizard occur locally. Surveys of the Site confirmed the previous records of grass snake within the Study Area, although only low numbers were observed. Grass snake are typically widespread in Wiltshire, albeit localised to suitable habitats. Past records of this mobile species from the general area are relatively numerous and as such support the experience of FPCR that this species is relatively widespread throughout suitable habitats associated with the River Cole, its tributaries and associated riparian and wetland habitats. This species is therefore also likely to be of local significance.

9.4.59 Further detailed survey information is provided in **Appendix 9.4.**

Breeding and Overwintering Birds

9.4.60 Breeding birds surveys completed within the Site in 2010 identified a total of 58…. species. This data included a number of notable species included within the UK BAP, Schedule 1 or that are Red Listed, and is summarised in Table 9.11 below:

Table 9.11 WCA Schedule 1, NERC S41 and BoCC red and amber listed bird species recorded in the Study Area during breeding bird surveys 2011 and their recent status within the county.

Species	Legal/ Conservation status	Maximum Count / Survey Occasions Recorded	Breeding Status†	Recent Status in Wiltshire
Bullfinch *Pyrrhula pyrrhula*	Amber List NERC S.41 LBAP	2 / 3	Probable	Moderately common resident of woods and scrub
Cuckoo *Cuculus canorus*	Red List NERC S.41 LBAP	1 / 1	Possible	Widespread but decreasing visitor from Africa
Dunnock *Prunella modularis*	Amber List NERC S.41	10 / 3	Probable	Still common and widespread resident, though decreasing Britain as a whole
Green Woodpecker *Picus viridis*	Amber List	1 / 2	Possible	Widespread resident, increasing Britain as a whole
Grey Partridge *Perdix perdix*	Red List NERC S.41 LBAP	2 / 1	Possible	Uncommon and Long decreasing resident, some released/naturalised stock

Species	Legal/ Conservation status	Maximum Count / Survey Occasions Recorded	Breeding Status†	Recent Status in Wiltshire
Herring Gull *Larus argentatus*	Red List NERC S.41 LBAP	2 / 2	Non-breeder	Scarce visitor from elsewhere in Britain/north Europe. Recent and increasing local breeder
House Martin *Delichon urbica*	Amber List LBAP	24 / 2	Possible	Common summer visitor/passage
House Sparrow *Passer domesticus*	Red List NERC S.41 LBAP	4 Colonies / 3	Probable	Common resident, trends in Wiltshire uncertain but large decrease nationally since 1970s
Kestrel *Falco tinunculus*	Amber List	1 / 1	Possible	Common but decreasing resident, also passage migrant and winter visitor
Lesser Black-backed Gull *Larus fuscus*	Amber List	3 / 1	Non-breeder	Increasing passage/winter from GB/Iceland/north Europe, recent and increasing local breeder
Linnet *Carduelis cannabina*	Red List NERC S.41 LBAP	8 / 3	Probable	Seriously decreasing resident, also passage migrant and winter visitor
Mistle Thrush *Turdus viscivorous*	Amber List	4 / 2	Probable	Common resident and perhaps partial migrant
Red Kite *Milvus milvus*	Schedule 1 Amber List	1 / 1	Non-breeder	Rare resident and scarce wanderer from Wales and re-established populations in England
Reed Bunting *Emberiza schoeniclus*	Amber List NERC S.41 LBAP	5 / 2	Probable	Seriously decreasing resident in wetlands, also some passage/winter
Skylark *Alauda arvensis*	Red List NERC S.41 LBAP	5 / 3	Probable	Still locally common resident/passage/winter, but serious declines in GB
Song Thrush *Turdus philomelos*	Red List NERC S.41 LBAP	5 / 3	Probable	Common resident, partial migrant, also winterer from north GB/Europe
Starling *Sturnus vulgaris*	Red List NERC S.41 LBAP	55 / 3	Probable	Seriously decreasing resident, common winter visitor from north Europe
Stock Dove *Columba oenas*	Amber List	7 / 3	Probable	Common and increasing resident of parkland, downland, wood edges, farmland with trees and barns, locally towns
Swallow *Hirundo rustica*	Amber List LBAP	47 / 3	Confirmed	Common and increasing summer/passage
Swift *Apus apus*	Amber List LBAP	5 / 2	Non-breeder	Common summer visitor

Species	Legal/ Conservation status	Maximum Count / Survey Occasions Recorded	Breeding Status†	Recent Status in Wiltshire
Whitethroat *Sylvia communis*	Amber List	24 / 2	Confirmed	Common summer visitor and passage migrant
Willow Warbler *Phylloscopus trochilus*	Amber List	15 /3	Probable	Common summer visitor
Yellow Wagtail *Motacilla flava*	Red List NERC S.41 LBAP	4 / 3	Confirmed	Local summer visitor
Yellowhammer *Emberiza citronella*	Red List NERC S.41 LBAP	9 / 3	Confirmed	Decreasing resident, though still widespread on arable farmland

9.4.61 Overwintering bird surveys recorded during winter 2012/13 identified a total of 53 species, many of which are either winter visitors or migrants only. Of the species recorded some, including barn owl, great black-backed gull, lesser redpoll and red kite are considered to be scarce or rare within the county. Nevertheless, given the size of the Site and the habitats present, the assemblage is broadly as could be expected and is considered to be of significance at a local scale.

Table 9.12 WCA Schedule 1, NERC S41 and BoCC red and amber listed bird species recorded in the Study Area during wintering bird surveys 2012-13 and their recent status within the county.

Species	Legal/ Conservation status	Maximum Count / Survey Occasions Recorded	Recent Status in Wiltshire
Barn Owl *Tyto alba*	Schedule 1 Amber List LBAP	1 / 1	Scarce but widespread resident, increasing again after serious decline
Black-headed Gull *Larus ridibundus*	Amber List	56 / 2	Local breeder, common winter/passage
Bullfinch *Pyrrhula pyrrhula*	NERC S.41 Amber List LBAP	6 / 3	Moderately common resident of woods and scrub, visiting gardens in winter
Dunnock *Prunella modularis*	NERC S.41 Amber List	18 /3	Common and widespread resident, decreasing in Britain
Fieldfare *Turdus pilaris*	Schedule 1 Red List	68 / 3	Common winter/passage

Species	Legal/ Conservation status	Maximum Count / Survey Occasions Recorded	Recent Status in Wiltshire
Great Black-backed Gull *Larus marinus*	Amber List	2 / 1	Scarce migrant/winter visitor, breeds coastal and north Europe, winters also inland in GB
Green Sandpiper *Tringa ochropus*	Schedule 1 Amber List	1 / 1	Uncommon passage migrant and winter visitor
Green Woodpecker *Picus viridis*	Amber List	2 / 1	Widespread resident, increasing throughout Britain
Grey Wagtail *Motacilla cinerea*	Amber List	1 / 1	Common winter visitor, also some passage/winter from north Britain
Herring Gull *Larus argentatus*	NERC S.41 Red List LBAP	70 / 3	Scarce winter visitor from elsewhere in Britain/north Europe, recent and increasing local breeder
House Sparrow *Passer domesticus*	NERC S.41 Red List LBAP	5 Colonies / 3	Common resident, trends in Wiltshire uncertain but decreasing nationally
Kestrel *Falco tinunculus*	Amber List	2 / 1	Common but decreasing resident, also passage migrant and winter visitor
Lesser Black-backed Gull *Larus fuscus*	Amber List	190 / 3	Increasing passage/winter from GB/Iceland/ north Europe, recent and increasing local breeder
Lesser Redpoll *Carduelis cabaret*	NERC S.41 Red List LBAP	4 / 1	Scarce winter visitor from north England, no recent evidence of breeding
Linnet *Carduelis cannabina*	NERC S.41 Red List LBAP	6 / 3	Seriously declining resident, also passage migrant and winter visitor
Little Egret *Egretta garzetta*	Amber List	1 / 1	Resident, but localised breeder
Mallard *Anas platyrhynchos*	Amber List	7 / 3	Common resident (some released) and winter visitor from Europe
Meadow Pipit *Anthus pratensis*	Amber List	6 / 3	Common resident, and passage/winter
Mistle Thrush *Turdus viscivorous*	Amber List	5 / 2	Common resident and perhaps partial migrant
Red Kite *Milvus milvus*	Schedule 1 Amber List	1 / 1	Rare resident and scarce wanderer from Wales and re-established populations in England
Redwing *Turdus iliacus*	Schedule 1 Red List LBAP	153 / 3	Common winter/passage

Species	Legal/ Conservation status	Maximum Count / Survey Occasions Recorded	Recent Status in Wiltshire
Reed Bunting *Emberiza schoeniclus*	NERC S.41 Amber List LBAP	12 / 3	Seriously decreasing resident in wetlands, also some passage/winter
Skylark *Alauda arvensis*	NERC S.41 Red List LBAP	33 / 3	Locally common resident/passage/winter, but serious decline in GB
Song Thrush *Turdus philomelos*	NERC S.41 Red List LBAP	9 / 3	Common resident, partial migrant, also winter from north GB/Europe
Starling *Sturnus vulgaris*	NERC S.41 Red List LBAP	189 / 3	Seriously declining resident, common winter visitor
Stock Dove *Columba oenas*	Amber List	1 / 1	Common and increasing resident
Woodcock *Scolopax rusticola*	Amber List	3 / 3	Decreasing resident, under recorded winter visitor from north/east Europe
Yellowhammer *Emberiza citronella*	NERC S.41 Red List LBAP	42 / 3	Decreasing resident, though still quite widespread on arable farms

9.4.62 Further detailed survey information is provided in **Appendix 9.5**.

Water Voles

9.4.63 This species was noted as being present just outside of the Site to the east on the South Marston Brook to its confluence of the River Cole Where this species was also noted. Evidence observed included latrines, dropping piles, burrows and prints; and was indicative of a healthy population. The habitat in this location was generally good with a well-developed marginal fringe of bank-side vegetation consisting of ruderals and tall herbs. Occasional stretches of emergent in stream vegetation consisted of branched bur-reed, reedmace and watercress. The population of this species is considered to be of local significance.

9.4.64 No evidence of this species was observed within the Site during the surveys conducted in 2010. Habitat within the Site was generally considered to be suboptimal habitat value due to the presence of over shading vegetation over the majority of its length.

Otters

9.4.65 Evidence that otters are present within the wider landscape was observed during surveys of the wider area in 2006, when spraints were noted in riparian habitats associated with the River Cole. This survey data is further supported by information provided by the Environment Agency (Daryl Buck, pers. comm) suggesting that otter road casualties had been recorded close to Acorn Bridge and further south on the A419. This previous information is further supported by surveys completed in 2013, which identified frequent spraints throughout the eastern extent of the River Cole as it flows beyond the Site.

9.4.66 No evidence of otters within the Site was noted, although their passage through the Site may occur given the often large range of the species, and it is quite probable that this species could disperse through the Site on an occasional basis. The Site also provides potentially suitable places for rest or shelter, including couches, hovers or holts, in the form of overhanging bank-side trees and roots which are prevalent along the South Marston Brook. However, no evidence of any place of rest of shelter was observed.

9.4.67 Although not considered to currently be of significant value to otters, the Site is considered to be of significance at a District scale for otters, which are likely to be recolonising new sites and increasing their range along many river catchments.

9.4.68 Further detailed water vole and otter survey information is provided in **Appendix 9.6**.

White-clawed crayfish

9.4.69 The presence of white-clawed crayfish has been confirmed from the South Marston Brook outside of the Site boundary and from the Lenta Brook to the south of the Site, south of the A420. A range of age classes were noted and the population appeared to be healthy, although at a low density. Habitat within the brook is generally variable, with large areas within the Site generally devoid of suitable refuges. However, a total of nine suitable habitat patches were surveyed and no evidence of crayfish observed. They are known to be present just outside of the site boundary and therefore could occur at a very low density such as to preclude detection by normal methods.

Badgers

9.4.70 The presence of badgers has been confirmed within the Site with a single main sett present (see separate confidential Badger Survey **Appendix 9.7)**. All arable land and areas currently used for turf production are of extremely limited value as they are likely to provide little potential foraging habitat. Similarly, coarse unmanaged semi-improved grassland habitats are also of limited value unless cut as this species tends to prefer shorter swards for foraging. More important habitats include the shorter grassland habitats, hedgerows and the wooded South Marston Brook corridor.

Summary of Valued Ecological Receptors

9.4.71 Features of Local/Low value or above are considered to represent a 'Valued Ecological Receptor' (VER) with potential to experience significant effects, should impacts be sufficiently adverse. Those features not meeting the criteria for VERs were classified as having either Site level or negligible ecological value.

9.4.72 A summary of VERs identified within or within the potential zone of influence of the Site are identified in Table 9.13 below.

Table 9.13: Summary value of Ecological Receptors

Valued Ecological Receptor	Nature Conservation Significance	Rationale
Designated Sites		
North Meadow and Clattinger Farm SAC	International/High	Site designated as a SAC and priority habitat listed under the NERC Act
River Cole CWS (including the South Marston Brook)	County/Medium	Designated as of county importance. Provides a relatively continuous network of semi-natural and less intensively managed habitats and potential dispersal habitat for protected species.
Other Habitats		
Broad-leaved plantation woodland	Local/Low	Some potential value and appreciably enriches the habitat diversity of the site
Ponds	Local/Low	Some potential value and appreciably enriches the habitat diversity of the site
Mature trees	Local/Low	Some potential value and appreciably enriches the habitat and structural diversity of the site

Valued Ecological Receptor	Nature Conservation Significance	Rationale
Veteran trees	District	Habitat of principal importance under S41 NERC Act and uncommon resource in the wider area
Hedgerows	Local/Low	Semi-natural/undisturbed habitat that is likely to provide a habitat for a range of wildlife and potentially important dispersal function through the Site
Fauna		
Great crested newts	District	GCN are protected under the Wildlife and Countryside Act 1981 and Habitats Regulations 2010 and are included on the NERC Act S41 list. Small but declining populations. Relatively widespread species generally although localised.
Bat assemblage	District	Supports a typical suite of species characteristic of farmland habitats. Relatively rare species recorded but uncommon and at low frequency All bats are protected under the Wildlife and Countryside Act 1981 and Habitats Regulations 2010 and are included on the NERC Act S41 list.
Reptiles	Local	Small population of grass snake which is partially protected and included on the NERC Act S41 list
Breeding and overwintering birds	Local	Supports a typical assemblage of breeding and over-wintering birds including the specially protected species barn owl and red kite.
Water vole	Local	Small population of water voles downstream, which is partially protected and included on the NERC Act S41 list
Otter	Local	Otter are protected under the Wildlife and Countryside Act 1981 and Habitats Regulations 2010 and are included on the NERC Act S41 list.
White clawed crayfish	County	Small population of white-clawed crayfish downstream, which as a species is partially protected and included on the NERC Act S41 list and declining species nationally
Badgers	Local	Badger social group present, which is protected under the Badger Act

9.5 POTENTIAL EFFECTS

9.5.1 The legislation requires that attention be paid to all likely forms of effects. These may be:

- Direct or indirect;
- Short or long-term;
- Intermittent, periodic or permanent;
- Cumulative.

9.5.2 Potential effects prior to mitigation include:

- Direct loss of habitats and associated flora and fauna within the Site boundary, interruption of wildlife corridors, decrease in value to wildlife through reduction in species and / or habitats;
- Indirect effects on retained vegetation within and bordering the Site, through increase disturbance and through local changes in soils, drainage and hydrology;
- Potential effects upon protected and scarce species through disturbance;
- Operational effects such as pollution incidents from chemical spills, pollution of streams and fragile habitats from runoff and incorrect storage of materials;
- Long-term beneficial effects arising as a result of the favourable habitat creation/enhancement on the Site and beneficial after-use.

9.5.3 From the outset and following review of the ecological baseline, the potential effects arising as a result of the design of the Proposed Development have been reviewed in order that, where possible, potential impacts can be avoided through an alteration in design, layout or working methods.

9.5.4 The assessment of potential impacts has been made with particular reference to:

- the description of development (Chapter 2 of this ES);
- the Development Framework Plan (**Appendix 2.1**); and
- the Green Infrastructure Framework- Guiding Principles (**Appendix 8.5**).

Construction Stage

Potential construction impacts on designated sites

9.5.5 No statutory designated sites of nature conservation significance would be affected by construction works.

9.5.6 The South Marston Brook, a part of the River Cole CWS - a non-statutory designated site - does however pass through the Application Site. In the absence of suitable mitigation, there is some potential for accidental pollution incidences and silt-laden construction site run-off to affect the Brook and habitats downstream within the wider River Cole CWS leading to an impact on both flora and fauna. The Brook supports particularly sensitive receptors such as the white-clawed crayfish. The effect on the CWS is likely to be only temporary in nature, although could if severe lead to effects of medium magnitude and a change in aquatic species composition.

9.5.7 Further effects will arise as a result of the implementation of two crossing points over the South Marston Brook, which are likely to lead to the loss/disturbance of stretches of c. 15 - 20m in length totalling up to 40m.

Potential impacts through habitat loss

9.5.8 Semi-natural habitats of significant nature conservation value are localised within the Site due to the predominance of intensive agriculture over much of the Site. As a result, the vast majority of built development (including some 65 ha of residential development, highway infrastructure and 2.3 ha of school and playing fields) will occur on arable or improved/species-poor semi-improved grassland of negligible nature conservation value.

9.5.9 Nevertheless, some small areas of value at a local scale are present including:

- broad leaved plantation woodland planted as a biofuel crop;
- mature trees located along the South Marston Brook corridor and along field boundaries;
- hedgerows distributed throughout the Site; and
- ponds scattered within the Site.

9.5.10 Wherever possible, these features have been retained within the Proposed Development as part of the iterative design process. However, some permanent effects could be expected as outlined below.

Broad-leaved plantation woodland

9.5.11 The proposals will lead to the loss (c 11.7ha) of a proportion of the young willow plantation intended to be used for biofuel. The magnitude of loss is considered to be

of Moderate significance as this habitat would still remain within the local area to the east and north east of the Application Site.

Hedgerows

9.5.12 The Site does support a network of hedgerows a number of which, due to their lack of intensive management and/or continuous nature, are of nature conservation interest in a Local context.

9.5.13 While the length of hedgerows lost has been minimised as part of the iterative design of the Proposed Development, some inevitable loss and disruption would be unavoidable. This includes:

- loss or partial loss of individual hedgerows (H11, H16, H27, H33, H34, H36, H60, H61, H62, H63, H72, H73, H75, H77, H78, H80 and H82); and
- The loss of a number of individual sections of c. 20m length,) arising as a result of the implementation of road/site access routes (H1, H4, H3, H9, H17, H18, H23, H25, H26, H29, H32, H35, H44, H69).

9.5.14 This represents only a small proportion of the overall hedgerow resource, which would be largely maintained within the Site's GI, and lead to a loss of moderate magnitude

Mature trees

9.5.15 The retention of mature trees has also been sought during the iterative design process. However, some unavoidable loss is likely to occur to provide access routes leading to effects on the tree resource of a minor magnitude. This includes:

- T23 – residential parcel
- T29 (category U) – residential parcel
- TG3 – New access roundabout at Thornhill Road.
- TG9 - Residential parcel and main street
- TG10– Main street
- TG18– Main street
- TG 24 – Main Street
- TG20– Main street

- T48 and T49– Main street
- TT47 and T50 – Main street
- T54– New access roundabout at Old Vicarage Lane
- T55 and T56– New access roundabout at Old Vicarage Lane
- T63– Upgrade to junction A420 and Old Vicarage Lane
- T90– Main street
- TG53– Residential parcel and street
- TG64– section of the group to open up for the street
- T181 – T184, TG69, TG70 and TG71 – Junction at the easternmost part of the Site meeting with the A420
- TG36 – section of group
- T135 – link road
- T169 – drainage basin

9.5.16 A further detailed review of tree losses is detailed within the Arboricultural Assessment at **Appendix 9.8**.

Effect of Habitat Loss and Disturbance on Fauna

Great crested newt

9.5.17 The presence of two GCN populations has been identified within the Site. Each is considered to be a distinct population isolated from each other by roads, the South Marston Brook and by distance. Both are also of small size and apparently declining and are therefore inherently vulnerable to local extinction following habitat or environmental change.

9.5.18 Whilst recorded in association with two ponds (P1 and P4), individual newts can use terrestrial habitats up to 500m from a breeding pond[9]. It is, however, generally accepted that where suitable habitat is present, the majority of any GCN population will use terrestrial habitats within 50 metres of the breeding pond (see for example, Jehle (2003))[10].

[9] English Nature (2001) GCN Mitigation Guidelines, Peterborough
[10] Jehle, R. (2000). The terrestrial summer habitat of radio tracked GCNs (*Triturus cristatus* and marbled newts (*Triturus marmoratus*). The Herpetological Journal 10:137-143

9.5.19 English Nature (now Natural England)'s *Great Crested Newt Mitigation Guidelines* (August 2001) state that suitable terrestrial habitats within 250 metres of a breeding pond are likely to be used most frequently by GCNs. More recent research commissioned by Natural England has shown that GCN densities are very low over 100m from the breeding pond, and that the majority of GCNs occur within 50m of the pond[11].

9.5.20 Natural England, in order to aid the identification and assessment of impact, has identified three terrestrial zones: immediate (land within 50m of the breeding pond), intermediate (land between 50 and 250m of a breeding pond) and distant habitat (land over 250m from the breeding pond), which can be used to quantify impacts based on the value of habitat.

9.5.21 The Proposed Development would lead to the loss of terrestrial habitat within the Site only and the majority of this is considered to be of sub-optimal nature being mainly arable in nature. Aquatic breeding habitat will be retained. A summary of habitat loss is provided in table 9.14 below:

Table 9.14: Summary of GCN habitat to be lost on Site

Receptor	**Habitat losses**		**Habitat to be retained and enhanced within Site**	
	Immediate terrestrial habitat (<50m from pond)	**Intermediate terrestrial habitat (50-250m from pond)**	**Immediate terrestrial habitat (<50m from pond)**	**Intermediate terrestrial habitat (50-250m from pond)**
Terrestrial habitat around P1 only	0ha	13.5ha (of which 2.2ha is school)	0.8ha	5.4ha
Terrestrial habitat around P4 only	0.2ha	9.2ha	0.6ha	9.7ha

9.5.22 In the absence of mitigation, the magnitude of terrestrial habitat loss would lead to the inevitable effect on the favourable conservation status of GCNs within the Site

[11] Creswell and Whitworth, (2004). An assessment of efficiency of capture techniques and the value of different habitats for the GCN. Herpetological Journal vol. 10, p 143-5

and, potentially, their killing and injury and disturbance of their places of rest or shelter during site clearance operations.

Bats

9.5.23 While the loss of trees is to be largely avoided, as discussed above some tree removal may inevitably be required for the provision of access routes through the Proposed Development. Trees with bat roost potential have been identified and are detailed in the Bat Survey Report (**Appendix 9.3**) and where these will be directly affected by proposals they have been aerially inspected to determine the status of bats. These identified a single occasionally used suspected noctule roost within tree T50. As this roost is likely to be one of many within the range of the local population and it does not perform an important function such as supporting a maternity colony its loss is not likely to be of any more than of minor significance.

9.5.24 Similarly, the proposals will also lead to the loss of a small occasionally used pipistrelle roost located within the stable of Rowborugh Farm. Only a single bat was observed to emerge of two occasions suggesting that the roost site is not of high significance for any local population. Its loss is therefore considered to be of minor significance to the local population.

9.5.25 Furthermore, additional roost habitat could be created through natural processes such as wind damage or rot and additional trees may require removal at a later date for health and safety reasons or as part of any future Reserved Matters applications. If bat roosts are confirmed as present, the impact is likely to be minor at most, as tree roosts are generally transitory in nature. However, a precautionary approach is adopted and it is fully expected that, were a bat roost discovered, appropriate mitigation could be easily incorporated into the proposals such that any impact is minimised for the species of bat likely to occur within the Site and identified during the desk study.

9.5.26 The most important habitats/areas for bats, such as those along the river/brook corridor and major associated hedgerows/woodland and around existing ponds, are to be largely retained and incorporated into the green infrastructure. Nevertheless the proposals will lead to the loss of some hedgerows that are used by bats and there will therefore be some alteration in the pattern of use of the site by bats. This is likely

to lead to some minor effect, although is unlikely to significantly affect any local population of any species identified.

9.5.27 The small losses of hedgerow to allow access (i.e. up to 20m stretches) are unlikely to affect bat corridor routes as these distances are regularly crossed by members of this group. However, the use of high intensity lighting can impact on bats during the construction stage, particularly where lighting is close to important habitat such as that used for foraging and as corridors of movement by some of the slower flying species such as the *Myotis* and brown long-eared bat species. While this is unlikely to significantly affect the viability of any local population, it could lead to their displacement from some areas or decline in foraging efficiency in the short-term during construction. Other species, including Noctule, perhaps the most notable species recorded and *Pipistrellus*, the most numerous, are likely to benefit from the increase in floodlighting as they feed on the insects attracted to the light.

Reptiles

9.5.28 Grass snakes were found to be present within the Site at low density. Suitable habitat for this species exists in association with edge habitats located throughout the Site which may be used for dispersal and along the South Marston Brook corridor where a mosaic of riparian habitats occurs and provides the appropriate juxtaposition of areas of cover, basking and hunting. Much of these important habitats would either be unaffected or will be enhanced for this species.

9.5.29 Nevertheless, the construction works will lead to the minor loss of reptile habitat, the displacement of reptiles and potential for direct mortality during initial ground works.

9.5.30 In the absence of mitigation, initial construction works could lead to the accidental killing of reptiles and, although unlikely to affect the anticipated wider population, impacts of Minor magnitude may occur.

9.5.31 All common reptile species thought likely to occur within the Site are partially protected under Schedule 5 (Sections 9(1) and 9(5)) of the Wildlife and Countryside Act 1981 (as amended) (Ref. 13.14). This legislation protects these animals from:

- reckless or intentional killing and injury; or
- selling, offering for sale, possessing or transporting for the purpose of the sale or publishing advertisements to buy or sell a protected species.

9.5.32 Where these animals are confirmed as present on land that is to be affected by development, guidance recommends that:

- The animals should be protected from injury or killing during construction operations;
- Mitigation should be provided to maintain the conservation status of the species locally.

Breeding and Over-wintering Birds

9.5.33 The Site was not considered to potentially provide any particularly valuable habitat for wintering or breeding bird species, predominantly due to the dominance of intensively managed arable and improved grassland habitat. As a result, it was not considered likely to support any particularly rare or vulnerable species. However, the Site is likely to provide a habitat for a range of common and widespread farmland bird species of conservation concern. Those that are more likely to be negatively affected by the proposals are those that have more restricted habitat requirements such as skylark and lapwing; ground nesting species that may opportunistically use habitats within the Site when crop type/management is conducive; and those species more tied to open farmland habitats, such as linnet and yellowhammer. However, as a result of the widespread nature of the majority of species recorded utilising habitats within the Site, a significant impact on any wider population is considered reasonably unlikely for the majority of species.

9.5.34 The effect of habitat loss on the breeding bird assemblage, which, given the nature of habitats on Site is not considered to be any greater than Local significance, is unlikely to lead to a major reduction in the species richness of the Site, although a reduction in the local population size of many of the farmland specialists could be expected leading to effects of Minor magnitude.

9.5.35 Construction works, such as initial ground and vegetation clearance, also have the potential to disturb breeding birds and potentially cause the loss or abandonment of nests. Whilst there is unlikely to be any major impact on local bird populations, it should be noted that all birds, their nests and eggs are protected by law and it is an offence, with certain exceptions, to intentionally:

- kill, injure or take any wild bird;

- take, damage or destroy the nest of any wild bird while it is in use or being built;
- take or destroy the egg of any wild bird; and
- in the case of birds listed on Schedule 1 (e.g. barn owl), disturb any bird while it is nest building or is at or near a nest with young; or disturb the dependant young of such a bird.

9.5.36 The potential for impacts on barn owl can be summarised as those associated with loss of a potential nest site and foraging habitat. Both features are considered to be limiting factors within the wider landscape, potentially reducing the further dispersal and colonisation. In the absence of mitigation, the loss of either could limit the success of this species at a local scale and its potential loss from the immediate area may arise.

9.5.37 Whilst proposals do not indicate the loss of the nest site, suitable foraging habitat is highly limited within the site. Ideal foraging habitat for barn owls support large numbers of field vole. The optimum habitat for field voles is rough tussocky grassland with a thick litter layer through which field voles will tunnel and nest. Arable crops are of limited value although grassland areas may be of more value prior to cutting for hay, which is currently the main agricultural practice undertaken within grassland field compartments in the eastern part of the site. Grassland habitats to the south of South Marston are of limited value. Although some limited habitat for small mammals is likely to occur along field boundaries, optimal field vole habitat is currently localised and is only present in a quantity between the Railline and willow coppice where approximately 2.4 ha occurs. Further good quality habitat occurs off-site to the north where at least 5.8ha is present.

9.5.38 Typically, during the breeding season most barn owl hunting occurs within a 1km radius of a nest site, in which there should be at least 4ha of rough grassland for breeding to be successful. Barn owls have been known to breed with much smaller areas available, with less than 0.5ha within 1km of the nest. As at least 5.8ha of suitable foraging habitat is potentially available, and with more anticipated to occur further afield, the loss of a small area of habitat within the site is unlikely to be of more than minor magnitude.

Badger

9.5.39 The presence of badger setts has been confirmed. However suitable foraging and corridor habitat is largely focused on hedgerows and on woodland, scrub and grassland habitats located in the north of the Site with little potential associated with intensively managed arable habitats.

9.5.40 Construction activities could lead to some loss of suitable foraging habitat arising as a result of the loss of species poor semi-improved grassland and hedgerows. However, a significant area of retained good quality habitat will remain throughout the Site. Badgers often range over large distances to forage with normal ranges extending to between 15ha and 300ha (Neal and Cheeseman 1996, Kruuk 1989) and it is generally accepted that the loss of 25% or more of the social groups range could have a significant negative impact. The site is, however, expected to represent only a small proportion of the badger social group's likely range and given the nature of habitats to be lost the loss of habitat is unlikely to lead to any reduction in the viability of the badger social group. Furthermore, in urban areas, food availability to badgers is considered to be often over-abundant suggesting that the loss in foraging habitat is only likely to be short-term in nature, until garden and sub-urban habitats develop. The impact on badgers through a loss of habitat is therefore considered to be short-term and of minor magnitude.

9.5.41 A badger sett has been identified within the site. Badger setts are protected from interference or disturbance under the Protection of Badgers Act 1992 and while badgers setts within the development are likely to be retained due to their location in hedgerows, some disturbance could occur. This is unlikely to significantly impact on badgers as it is generally considered that they are relatively tolerant of disturbance, given the fact that they regularly occupy area where disturbance appears high. Nevertheless, where construction works occurs close to setts there is some potential for setts to be damaged, specifically subterranean parts. Whilst this is unlikely to significantly affect the badger social group (due to the main setts locations off-site), in the absence of mitigation this could lead to an effect on individual badgers, potentially causing entrapment and leading to an offence under the above act and a minor negative effect.

Potential Construction Effects on other Fauna

9.5.42 The presence of a number of additional species, including water vole, otter and white-clawed crayfish, has been identified in association with the South Marston Brook. Direct effects on these species during construction are unlikely. However, white-clawed crayfish could be affected by any effects on water quality as discussed in paragraph 9.5.6. If severe, the impact could lead to population decline and potentially local extinction from habitat immediately downstream of the Site.

Fragmentation and Isolation of Habitats

9.5.43 The development layout and green infrastructure provision has sought to provide links from adjacent corridor habitats, such as the woodland habitats on the eastern boundary, into the retained parts of the Site. The Proposed Development will lead to the loss of a number of hedgerows and the implementation of some features within the proposals, such as the access routes within the development, has the potential to further isolate and fragment existing habitats. This could slow the colonisation of species and limit the use of the occupied Site by wildlife. Valued ecological receptors that could be affected in the absence of mitigation include potentially bats, water vole and otter, leading to a reduction in the use of the Site and effects of up to Moderate magnitude.

Potential Degradation through Pollution

9.5.44 In the absence of suitable mitigation, there is some potential for accidental pollution incidences and silt-laden construction site run-off to affect retained wetland features and habitats downstream, such as the River Cole, leading to a Minor impact on both flora and fauna.

9.5.45 Additional impacts can arise as a result of an increase in airborne dust, during periods of dry weather when soil-stripping and earth moving is being undertaken for example. This can damage vegetation and potentially affect associated fauna including invertebrates, which are particularly susceptible. Where impacts are severe, some species may disperse from affected areas in the short-term. This is therefore likely to lead to a short-term negative impact on lower plants and invertebrates and is greatly dependant on mitigation measures and weather conditions. However, in the absence of mitigation, the impact from dust could lead to some Minor temporary effects.

Post- Completion Stage

Designated Sites

9.5.46 Potential effects on internationally designated sites are unlikely. In particular, consideration was given to the potential effects on North Meadow and Clattinger Farm SAC, which are further detailed in the Habitats Regulations Assessment Report for the *Swindon Local Plan 2026: Submission Version* (2013).

9.5.47 The screening work identified three main areas of potential impact (water resources, water quality and air pollution). The Appropriate Assessment concluded:

"The analysis shows, that the indirect pressures on water resources and water quality which may lead to likely significant effects at European Sites, arise from the predicted growth in demand resulting from new development; which in part arises from Swindon's Core Strategy. **These issues are, however, addressed systematically through water resource planning and in particular, through the regulatory assessments and augmentation/ management measures required by the EA in response to the Habitats Directive and the Water Framework Directive**. The potential for in-combination effects, which are most significant in relation to other developments driven by neighbouring LDFs are also considered by these processes."

"Air pollution in particular from NOx gases is a key concern if a road carrying a significant proportion of new traffic related to the plan runs within 200 meters of a European site." ……."the North Meadow and Clattinger Farm SAC is within 200m of the A419. However, evidence from current condition assessments at North Meadow and Clattinger Farm SAC indicates that air pollution is not having an adverse effect on the site and that site level management is the key factor in maintaining site integrity. **The plan is therefore unlikely to have an adverse effect on this site either alone or in-combination with other plans."**

[Emphasis given by FPCR only]

9.5.48 The above assessment concludes that the allocation of the site is unlikely to lead to likely significant effects on this site.

Disturbance to Wildlife

9.5.49 Operational impacts include the increased disturbance of wildlife resulting from the occupation and use of the Site by residents.

9.5.50 With an increase in domestic animals it would be expected that wildlife, including birds and small mammals associated with retained habitats, might be subject to increased predation and disturbance from cats and dogs. While this impact may affect the local distribution of some species, it is considered to be of Minor magnitude, being offset by the substantial increase in garden habitats which will occur, alongside structural landscaping.

9.5.51 Both birds and reptiles are considered likely to be affected by an increase in use of the area by humans. These species are likely to become habituated to some effects, such as noise from Site operations, although where the frequency is sporadic and the amplitude high, some avoidance or flight from it may occur. Birds are especially susceptible to noise from roads, although due to the anticipated level of use of on-site roads this is not likely to be significant.

9.5.52 The unmitigated lighting of any inter-connected habitats both within and adjacent to the Proposed Development, such as hedgerows, trees or woodland edge, may indirectly impact upon the faunal species that are dependent on them. Some bat species are known to avoid illuminated areas, which may lead to reduced foraging success and survival rates. Furthermore, the lighting of favoured foraging areas is likely to increase the risk of bat mortality from predation (*Bat Conservation Trust 2009,* Jones 2000). In the absence of mitigation, the direct lighting of retained habitats may reasonably be expected to the lead to the avoidance of these areas by sensitive species. Therefore, the impact of unmitigated lighting is considered likely to result in a minor-moderate impact to a receptor of local value where unmitigated.

Degradation of habitats from accidental pollution incidences and site surface water runoff

9.5.53 The discharge volumes expected from the Site are not considered to alter significantly from the existing greenfield run-off and all discharges will follow

guidelines produced by the Environment Agency. As such, the impact on habitats is unlikely to be significant.

9.6 MITIGATION MEASURES

9.6.1 From the outset and following review of the ecological baseline, the potential effects arising as a result of the design of the Proposed Development have been reviewed in order that, where possible, potential impacts can be avoided through an alteration in design, layout or working methods.

Construction stage

General Precautions

9.6.2 The potential for impacts on retained habitats outside of the immediate working area during construction activities would be minimised by retaining and protecting all unaffected habitats within the Site to ensure that disturbance is kept to a minimum and any sensitive species are retained. A buffer zone would be retained either side of all retained hedgerows. Depending on situation, a range of measures would be used in order to incorporate these features into the development. Similarly, works beneath the canopy/within the Root Protection Area (RPA) of any retained trees would be avoided or, if necessary, mitigated (refer to Arboricultural Assessment: FPCR, 2013; **Appendix 9.8**).

9.6.3 Both remediation and construction works required for the formation of development plots occur in fairly close proximity to the South Marston Brook. In order to prevent indirect effects through an alteration in water chemistry or siltation and the consequent smothering of flora and fauna, appropriate measures will be implemented in line with best practice.

9.6.4 The relevant EA Pollution Prevention Guidelines listed below will be adhered to, to ensure construction works are undertaken in an environmentally responsible manner. Any environmentally hazardous material used will be kept in dedicated stores and storage tanks will have appropriate bunding:
PPG1: General Guide to the Prevention of Pollution;
PPG2: Above Ground Oil Storage Tanks;
PPG3: Use and Design of Oil Separators in Surface Water Drainage Systems;
PPG5: Works in, Near, or Liable to Affect Watercourses;
PPG6: Working at Construction and Demolition Sites;
PPG21: Pollution Incident Response Planning;

PPG23: Maintenance of Structures over Water.

9.6.5 Earthworks will be undertaken according to the guidelines given in PPG6: Working at Construction and Demolition Sites. The following measures will be implemented as required to prevent pollution from earthworks:

- erosion control measures – including avoidance of works close to water bodies/South Marston Brook during inclement weather and covering of temporary stockpiles where necessary to prevent runoff from flowing across exposed ground and becoming polluted with sediments;
- sediment control measures – including cut-off ditches and silt traps as required to slow runoff and allow for settlement of sediment as close to the source as possible.

9.6.6 The loss of small sections of hedgerow, plantation woodland and established scrub habitats would be easily compensated for through the sensitive planting of appropriate species and subsequent management of GI habitats. Further detail regarding both the creation and management of GI habitats is provided below and presented in the Green Infrastructure Framework – Guiding Principles (**Appendix 8.5**).

9.6.7 Habitat creation and enhancement opportunities presented within the GI are driven by a number of factors including:

- a requirement to compensate for loss of habitat used by fauna and where possible ensure that there is a net gain in biodiversity arising as a result of the development
- the maintenance of habitat corridors through the Site and incorporation of habitat links to surrounding habitats of interest; and
- the presence of existing features of interest that would benefit through sympathetic management.

9.6.8 Particular consideration has also been given to the *Swindon Biodiversity Action Plan 2010*, *A Landscape Scale Framework for Conservation in Wiltshire and Swindon* (Wiltshire Wildlife Trust, 2012) and, in particular, the Area 02 – Swindon Ridge and Clay Lowlands section of that document and *A Green Infrastructure Framework for Swindon 2010-2026 (2011)*.

9.6.9 The proposals include the provision of an extensive Green Infrastructure (GI) package, totalling some 70ha or approximately 40% of the Application Site. The GI can be roughly subdivided into the following elements, from the most strategic to those of more localised significance:

- Rowborough Park – informal and formal open space extending along the South Marston Brook corridor;
- Nightingale Park – informal and formal open space extending along the railway corridor and linking with the River Cole and Nightingale Wood to the north;
- South Marston Park – located to the south of South Marston village; and
- All Other Greenspace and Development land –situated along existing field boundaries and ditches and within development parcels.

9.6.10 The Proposed Development would include a significant area of informal open space, thereby providing an opportunity to help the local population have access to and value a wildlife rich environment close to where they live and work. This is provided by an interlinking green infrastructure system having the South Marston Brook and Rowborough Park as its spine. Open land will be sustainably managed for people and wildlife; new built development will play a part in delivering new and enhanced habitats.

9.6.11 The following habitats will be created as part of the proposals and should be read in conjunction with the Green Infrastructure Framework (GIF) supporting the planning application. Guiding principles for the GIF have been based on *A Green Infrastructure Strategy for Swindon 2010-2026 (2011).*

Woodland, Scrub and Hedgerows

9.6.12 New hedgerows and scrub habitat would be created within public open space and within the GI, which provide a habitat corridor through the Site and wider area, linking with off-site semi-natural habitats in the northeast and east. Shrub species should be based on the composition of existing hedgerows bounding and within the Site. Retained hedgerows will provide sheltered corridors for wildlife to move along, encouraging dispersal throughout the Site. Hedgerows already provide good structural diversity and associated under-storey vegetation adds more variety to the habitat. This may be sustained through appropriate management with the aim of maintaining continuous hedgerows 3-4m wide.

9.6.13 Hedgerows will be trimmed regularly potentially every 5-10 years or perhaps on alternate sides every 2-5 years which would allow hedgerows to thicken; this may be combined with gapping up with similar species where necessary. Hedgerow management would be carried out during late winter to allow wildlife to take advantage of fruits and berries during the winter and during a period when trees are dormant and work will cause minimum disturbance to wildlife.

9.6.14 New woodland planting will be undertaken in order to augment and complement existing woodlands that exist locally such as Nightingale Wood to the north east and increase connectivity between them.. The planting mix would be representative of local woodlands. Little management would be undertaken initially to ensure that the area is as little disturbed as possible although some thinning/coppicing/pollarding may be required in time to encourage the development of structurally diverse woodland.

9.6.15 Where new tree and shrub planting within the structural landscaping and GI is to be undertaken, native species will be used to provide a habitat to a range of species-specific and phytophagous species. The mix will comprise at least 50% climax species such as oak and a range of pioneer tree and understory shrub species.

Species-rich grassland

9.6.16 Species-rich areas of grassland will be located widely within the Site in association with woodland and scrub habitats along greenways and more extensively in Rowborough Park, Nightingale Park, South Marston Park and built development areas, where intensive use and more formal management is not proposed.

9.6.17 It is also the intention to create a network of small glades and areas of more open grassland among areas of more developed habitat such as around the existing ponds in particular, which are linked by a network of greenways. This will provide the structural and microhabitat diversity required by a range of species including a number of the invertebrates, grass snake and scrub specialist birds known to be present locally or recorded within the Site, as well as amphibians including the great crested newt.

9.6.18 Grassland habitats would be created in areas currently intensively managed for agriculture. Where possible, topsoils would be removed or mixed with subsoils to provide a substrate more conducive to the development of a more species-rich sward. Ideally and in more extensive areas of open space, a variation in topography would be created with shallow banks and undulations left to provide a high level of variation in micro-topography.

9.6.19 Coarse grassland areas to be created will also be allowed to develop on more developed soils to allow vigorous growth and the development of a structurally diverse habitat suitable for the colonisation of a range of species.

9.6.20 It is likely that the seeding of grassland habitats will be required in order to prevent the establishment of weed or other undesirable species. A neutral grassland seed mix of native provenance will therefore be used.

9.6.21 Initial management of grassland areas is likely to be up to four cuts in the first year to reduce colonisation and competition from undesirable species, followed by cutting in autumn (only a third cut in any one year), to maintain a varied sward with frequent patches of short (<50mm) grassland.

Wetland

9.6.22 A variety of wetland features will be included within the Site, both to benefit biodiversity but also to attenuate surface water flows.

9.6.23 Additional ponds will be formed close to the existing ponds P1 and P4 to provide additional potential aquatic habitat for existing amphibian populations and as indicated in the Green Infrastructure Framework (GIF) - Guiding Principles.

9.6.24 The waterbodies will be shaped to provide a range of bank angles and heights. Gradients will vary from 15º - 25º from horizontal and will be enhanced by the excavation of small embayments. This will create differing conditions of light and temperature and will thus encourage diversification in the flora and associated fauna. Variations in water depth will be created from shallows to deeper pools (50mm – 1000mm) to enhance nature conservation value and to maximise the development of biodiversity. The shallowest areas will grade into an expanse of seasonally wet mud

that may attract a variety of invertebrates and plants, which will in turn attract other fauna including birds and mammals.

9.6.25 A significant opportunity also exists in association with surface water attenuation proposals to provide a range of habitats that would complement those of existing interest along the South Marston Brook. The primary function of the surface water drainage system will be to control flows and reduce the potential for pollution of the aquatic environment. However, It is important that SUDS are considered holistically and that each element is recognised as part of a network of habitats and wildlife corridors.

9.6.26 Principles followed will include:

- Where possible, SUDS will be located close to existing wetland areas, so plants and animals can naturally colonise the new SUDS ponds;
- Well vegetated shallow bays will be created allowing the establishment of areas of marsh;
- Smoothly finished surfaces would be avoided - although they give the impression of tidiness, they provide less micro habitat diversity for plants and animals; and
- If planting is essential, such as for the earlier phases, only native plants of local origin will be used.

Management Plan

9.6.27 All habitats created within the Application Site would be subject to a habitat creation/management plan produced in consultation. Furthermore, all created and retained habitats within the Site would be managed sympathetically for the benefit of biodiversity.

Great crested newts

9.6.28 Due to the clearance of terrestrial habitats that could be used as resting/hibernation places by GCNs, it is considered that a Natural England European Protected Species (EPS) licence will need to be secured for the proposals to proceed. A detailed mitigation strategy (Method Statement) will also need to be produced in support of the licence application. The mitigation approach will be to avoid killing or injuring

GCNs and to minimise the impact on the GCN population in order to maintain their 'Favourable Conservation Status' in their natural range.

9.6.29 The emphasis of mitigation will be to capture any newts present within the development zones, and relocate them to adjacent receptor areas, which form a part of the GI and which includes their aquatic breeding habitat and enhanced terrestrial habitats formed where habitats are currently unsuitable (i.e. arable land), which have been specifically created for that purpose.

9.6.30 Enhancements in this area will include the planting of native scrub species, restoration of existing ponds to provide permanent open water and provision of hibernacula and refuge piles. Additional works required include the clearance/reduction in over-shading by vegetation/trees and in the case of pond P1 the gradual removal of the coniferous plantation and restocking with broad-leaved species. Additional measures include the removal of choking vegetation where necessary to provide no more than 80% vegetation cover, the removal of dense over-shading vegetation to no more than 50% cover and the removal of silt from ponds suffering from early drying.

9.6.31 Where grassland is currently present in retained areas, the existing management regime will be altered, from relatively intensive agriculture to more sympathetic/relaxed management, to improve the botanical diversity and structure for amphibians.

9.6.32 In addition, two hibernacula would be provided close to each of the identified breeding ponds. These hibernacula would follow the specifications as detailed within the Great Crested Newt Mitigation Guidelines (English Nature 2001) and would provide additional areas for shelter or refuge on Site for GCN.

9.6.33 Following the enhancement of receptor habitats, trapping and translocation of GCN from development zones will be undertaken. This will only be carried out during the newt active season, ideally during the breeding migration period, catching newts as they move to the breeding pond in spring.

9.6.34 Temporary amphibian fencing will be installed within the development zones, with a particular focus on suitable habitat features present within 500m of the breeding

pond. These areas will then be trapped out as specified in the Great Crested Newt Mitigation Guidelines for a small population (i.e. minimum of 30 nights trapping).

9.6.35 Further measures may also be required depending on the detailed design of the site, these would be dealt with at the detailed design stage and could be expected to include measures to integrate built development with GCN mitigation. Such measures may include appropriate design of suds, drainage and roads, where close to GCN population centres.

9.6.36 Further detail is contained within **Appendix 9.2**.

Bats

9.6.37 The loss of small occasionally used roost site for both noctule and pipistrelle bats will occur during construction. Mitigation proposals will include the provision of a total of 90 bat boxes suitable for use by both species (in addition to most others known to occur locally), 40 bat tubes on built structures and 5 bat garages specially designed to be accessible to and used by roosting bats. The location of bats roost features would be best identified at the detailed design stage.

9.6.38 During the post-completion stage, gaps of small lengths of hedgerows to allow access are proposed. These hedgerows crossing the Site provide connective corridors through the Site. In order to maintain this function, all retained hedgerows will be maintained appropriately, which in addition to the retention of the majority of trees is likely to reduce the magnitude of any impacts.

9.6.39 To further minimise the potential impact on foraging routes when the hedgerow linkages within the developed Site are broken, the retained hedgerows will be reinforced with native species planting. Where the proposed breach of a hedgerow exceeds 10-15 m in length, the planting may also include the implementation of standard trees adjacent to the road/footpath which will grow to be above the level of vehicle movement. The lower branches of such trees should be regularly pruned back to the trunk in order to ensure that the most suitable flight line is above the maximum traffic height (where applicable low-level lighting columns may also be used in this instance to reduce the likelihood of the bats using the lower tree regions). The implementation of such ‘hop-overs’ will allow continued echolocation across the break thereby allowing continued usage of the hedgerow as a

foraging/commuting area. It will also reduce the potential for road traffic accidents to bats (and also to birds).

9.6.40 Whilst the hop-overs will take a little while to establish, the tree standards to be used shall be of an appropriate size and will be planted early in the life of the development. Whilst the breach will be present during the construction period until the hop-over is established, due to the small size of the proposed breaches the impacts upon bats potentially commuting along them is considered to be minor.

9.6.41 The other key impact considered was the potential effect of lighting across the Site. It is considered that the scale of this potential impact will be significantly reduced through the use of low-level light or directional lighting where possible and appropriate. It is recommended that Street- and flood-lighting also use directional LED lighting where possible instead of mercury or metal halide lamps as they emit limited levels of UV light (which attracts insects, and hence, bats) and the light is easily diverted. The lighting recommended will minimise light spill onto the potential foraging routes (including the proposed flood attenuation facilities) and minimise potential disturbance caused through lighting of these corridors of movement. Lighting levels would be as low as guidelines permit and only used where required for public safety. Where possible it is also recommended that lighting is timed (i.e. switched off during periods of limited vehicular usage) in order to further reduce any potential impact on bats.

9.6.42 The proposals lead to the loss of a number of hedgerows and the works to others as a result of the implementation of internal access routes. Nevertheless, large areas of habitat currently considered unsuitable for bat commuting or foraging (including arable fields and semi-improved grassland) will be enhanced as a result of the proposals as they will be incorporated into the GI proposals and managed less intensively with a focus on maximising biodiversity.

Reptiles

9.6.43 The presence of grass snake within the Site is well recorded, although the population appears to be at a low density. Nevertheless this species could be harmed during Site clearance works. Areas of greater suitability occur scattered throughout the Site where more structural diversity in the vegetation occurs with frequent patches of bare ground, loose rubble and close juxtaposition of areas of shelter and basking.

9.6.44 The removal of reptiles from working areas will utilise a range of methods based on the suitability of the habitat to be affected and season:

- Clearance of vegetation where possible over winter, where no suitable over-wintering/hibernacula habitat occurs
- Passive displacement, where clearance of small isolated areas of homogenous vegetation or suboptimal habitat is proposed
- Trapping and translocation - where extensive habitat occurs and where the likelihood of passive displacement is lower.

Avifauna

9.6.45 To avoid disturbance to breeding birds, where vegetation is to be removed it will generally be removed prior to the bird-breeding season (March to August/Sept). However, if this is not possible, vegetation will be checked prior to removal by an experienced ecologist. If active nests are found, vegetation will be left untouched until all birds have fledged. Specific advice will be provided prior to undertaking the clearance. This would be a statutory requirement due to the protection of all nesting birds and their nests under the Wildlife and Countryside Act, 1981. A suitably qualified ecologist would supervise this.

9.6.46 If an active bird nest is found in vegetation or other features/habitat to be removed, the area would be retained and clearance operations would recommence only when the nest is no longer in use.

9.6.47 Although the proposals are buffer from and will not lead to the loss of any barn owl nest site, at least four barn owl boxes will be erected in peripheral parts of the site as far from works as possible (refer to: The Barn Owl Trust advice leaflets: Nest boxes for use of vertical trunks leaflet no. 212 for design). Nest box siting would be agreed with the LPA prior to erection, although would follow the recommendations within the above leaflet.

9.6.48 Habitat enhancement will also be undertaken in the GI. This will include areas of rough grassland occupying up to a 1/3rd of the grassland present and new

[12] http://www.barnowltrust.org.uk/content_images/pdf/Nestboxes_for_use_on_Trees1357645529.pdf

woodland/scrub planting, which provides good foraging habitat in the early years of development. It is likely that on this basis the availability of foraging habitat locally would be significantly increased.

Water voles and Otter

9.6.49 The presence of both water vole and otter has been identified locally, although not from within the Site. It is therefore possible that both species could become active within the Site in the future. All activities that may disturb water voles (works within 5m of the South Marston Brook and its tributary) will be subject to a survey immediately prior to works to determine the presence/absence of water vole burrows, which may have been created since the initial survey detailed within this document.

9.6.50 In order to ensure that their future dispersal is not hampered by the Proposed Development, culverts will be designed to ensure that wildlife is able to pass underneath them during flood events. Free passage would be by means of either a specially provided tunnel or ledge located above flood water level. Design will be agreed with the Environment Agency prior to construction. However, it could be expected that ledges will be constructed of either solid concrete or wood and metal bolted onto the culvert and will be at least 500 mm wide, 150 mm above flood water level, with at least 600 mm headroom and accessible from the adjacent banks by ramps. Ramps will also be provided from the water, where appropriate.

Post Completion Stage

Habitats

9.6.51 The effects of increased visitor pressure on retained habitats could be detrimental in the absence of mitigation. This is especially important in consideration of the effects of trampling and soil compaction on grassland and woodland flora, which can be susceptible to trampling. However, no particularly sensitive species have been identified and many of the impacts associated with the use of the green infrastructure habitats can be mitigated by the operation of appropriate site management and zoning of areas; from relative seclusion to more intensive recreational use. The impact is therefore considered to be of minor significance and mainly focused on woodland and riparian habitats along the South Marston Brook and River Cole.

9.6.52 The layout of the Proposed Development has sought to provide links from adjacent corridor habitats, such as from the railway corridor, into the retained parts of the site along the South Marston Brook, and many species will therefore benefit from the green infrastructure proposals. Particular benefit can be realised where habitat creation/enhancement proposals create further complementary habitats and enhance those already present. However, some operational impact may occur through the severing of corridor habitats through the use of street lighting or through the vehicular use of the Site and through the disturbance of the GI by both occupants and their pets.

Reptiles

9.6.53 With an increase in domestic animals associated with residential development, it would be expected that wildlife, including grass snake, birds and small mammals associated with retained habitats, might be subject to increased predation and disturbance from cats and dogs. While this impact may affect the local distribution of some species it is considered to be of minor significance, offset by the substantial increase in other non-agricultural habitats which will occur, alongside structural landscaping. The impacts arising could not realistically be mitigated with certainty, thus leading to minor impacts overall.

Bats

9.6.54 Any lighting around retained habitat used by bats as corridor or foraging habitat can lead to a reduction in use by this group. Some species such as Myotis species, which are generally slower flying and are known to occur in small numbers within the Site, are particularly sensitive to lighting and will avoid heavily lit areas as a predator avoidance strategy. Given the small numbers observed, this is unlikely to significantly affect any local population, although some change in the use of the Site could occur and a minor impact could be expected.

9.6.55 To avoid any impacts associated with operational light spill to potential bat flight-lines or foraging habitat, the following methods will be recommended to prevent impacts:

- The direct lighting of existing trees and hedgerows or proposed flood attenuation facilities would be avoided
- Street- and flood-lighting should seek to use low pressure sodium or high pressure sodium instead of mercury or metal halide lamps,

- Lighting should be directional and light spillage avoided
- Lighting columns would in general be as short as possible, although in some locations taller columns would allow reduced horizontal spill
- Where appropriate and possible within the relevant guidance consideration should be given to reduced lighting levels
- Where road safety guidelines allow, roads would remain unlit where important hedgerows exist to provide unlit stretches

Birds

9.6.56 During the operational phase, in addition to the retained habitats, habitat creation proposals associated with the GI are proposed that will enhance the retained habitat for most bird species using the site. New and enhanced habitats are proposed including native hedgerow and woodland planting, pond creation and enhancement and species-rich grassland creation that will provide new nesting and foraging opportunities in time. In addition, areas of public open space created a part of the development will also provide a range of nesting and foraging opportunities within the development. For species that typically nest in buildings, nest boxes or nesting features will be incorporated onto new buildings to provide nesting opportunities. Further nest boxes would also be provided along the river corridor and in retained woodland habitat.

9.6.57 In common with the effect on grass snake noted above, the increase in domestic animals, particularly cat, may lead to an effect on small bird populations. Current research is inconclusive as to the actual effect that domestic cats can have on wild populations, although it is highly likely that in combination with a change in land use that the assemblage will change. Some species (e.g. Starling, finches and thrushes) are likely to benefit from any increase in available nesting habitat as a result of the implementation of landscape design proposals, although it is likely that those of open habitats (e.g. yellow wagtail and skylark) would avoid the site. Nevertheless, neither the species richness or number of birds using the site is likely to decline and the effect is not considered to be of any greater than minor significance.

Badgers

9.6.58 Badger densities within urban habitats have been shown to be similar to those in rural areas and established urban habitats are generally accepted to of, as such, any

long-term impacts arising as a result of the change in land use on the continued viability on the local badger population are generally unlikely. While the impact from loss of habitat is unlikely to significantly affect the viability of the badger social group centred to the west of the site, some enhancement to existing habitat could be expected as a result of the GI proposals, particularly where existing arable habitats are replaced by woodland and permanent grassland habitats, which are of greater value to foraging badgers.

9.6.59 The early provision of green infrastructure habitats will be undertaken and include significant areas of scrub, woodland and grassland associated with the GI. These habitat types would provide an increased foraging resource, which, with early establishment and appropriate management, would provide a minor benefit the badger social group in the medium and long-term.

9.6.60 In order to further minimise the potential for operational impacts to badger setts access to and around retained setts will be minimised through for instance the creation of buffer zones around these areas and/or dense thickets of thorny shrubs to deter access.

9.6.61 The severance and disruption of corridor routes, in the absence of mitigation, lead to increased badger road casualties potentially leading to a decline in the badger social group over time. This is unlikely to lead to the loss of the social group, although could lead to a reduction in its success. Operational impacts associated with any potential increase in badger road casualties would be minimised though the incorporation of wildlife tunnels corresponding to the existing field edge location. Tunnels would measure 600mm in diameter and incorporate the use of badger proof fencing to encourage badgers to use the tunnels. Proposals would comply with those detailed in the Design Manual for Roads and Bridges.

9.6.62 While some future badger access into residential habitats would inevitably occur and could not be realistically prevented, the potential for badger road casualties would be reduced due to the controlled traffic speeds associated with internal access routes.

9.7 RESIDUAL EFFECTS

9.7.1 The assessment of residual effects takes into account the mitigation measures outlined. Many of these measures are incorporated as an integral part of the design of the Proposed Development. This iterative process has resulted in the Proposed Development being designed and modified to take account of the surveys and assessments undertaken. This has enabled the extent and scale of the potential adverse effects to be continually appraised as part of the evolving Development Framework Plan and scheme design.

9.7.2 The design approaches adopted have included measures to avoid or reduce potentially significant adverse effects arising from the Proposed Development. Compensatory measures are also proposed where impacts, such as habitat loss, are unavoidable and where this is proposed, compensatory habitat creation measures have sought to provide habitats that complement those located within and close to the Site.

9.7.3 A summary of residual impacts resulting from construction and following the completion of the Proposed Development is presented in Table 9.15.

Construction stage

9.7.4 Residual effects arising during the construction stage relate principally to habitat loss. The loss of hedgerows during the construction stage and the resulting effect on the ability for wildlife to disperse through the Site and the wider area is unlikely to be fully compensated by the implementation of the GI.

Post-completion stage

9.7.5 Post-completion effects are largely associated with the potential disturbance of wildlife, through the introduction of new residents and/or their pets. This could lead to minor adverse effects with some species, such as some farmland birds and mammals avoiding the most heavily disturbed parts of the site.

Summary of effects

9.7.6 The effects identified are summarised in Table 9.15:

Table 9.15: Summary of effects

Potential effect	Significance (pre-mitigation)	Mitigation measure	Significance of residual effect
Construction stage			
Potential for pollution of South Marston Brook through construction site run-off pollution events	**Moderate adverse**	Sound construction site management, including: • Adherence to Pollution Prevention Guidelines • Retention and protection of buffer strips	**Not significant**
Habitat loss through implementation of South Marston Brook crossing points	**Minor Adverse**	Careful construction site methods including adherence to minimum working areas Sympathetic management of retained South Marston Brook habitats	**Minor Adverse**
Loss of young willow biofuel coppice	**Major Adverse**	Habitat creation proposals throughout GI network to include woodland scrub and wetland	**Minor Adverse**
Loss of hedgerow	**Moderate adverse**	Habitat creation proposals throughout GI network to include hedgerows, woodland, scrub and wetland	**Minor Adverse**
Loss of mature trees	**Minor Adverse**	Habitat creation proposals throughout GI network to include Hedgerows, woodland, scrub and wetland	**Short/medium-term Minor Adverse** **Long-term insignificant**
Loss of habitat for great crested newt	**Moderate adverse**	Habitat creation proposals to include the provision of woodland and scrub habitats within immediate and intermediate habitats and enhancement through hibernacula and refuge creation. Breeding pond enhancement through removing overshading vegetation.	**Not significant**

Disturbance/harm to great crested newt during site works	**Moderate Adverse**	Precautionary measures during site clearance/pre-commencement trapping and translocation under licence	**Not significant**
Loss of foraging/dispersal habitat for bats	**Minor Adverse**	Habitat creation proposals throughout GI network to include hedgerows, woodland, scrub and wetland	**Not significant**
Loss of roost habitat for bats	**Minor Adverse**	Provision of alternative roost habitat on trees and buildings (following construction)	**Not significant**
Loss of habitat for reptiles	**Minor Adverse**	Habitat creation proposals throughout GI network to include hedgerows, woodland, scrub and wetland	**Not significant**
Disturbance/harm to grass snake during site works	**Moderate Adverse**	Precautionary measures during site clearance/pre-commencement trapping and translocation	**Not significant**
Loss of habitat for breeding and overwintering bird assemblage	**Minor Adverse**	Habitat creation proposals throughout GI network to include hedgerows, woodland, scrub and wetland	**Minor Adverse**
Disturbance of nesting birds	**Minor Adverse**	Avoidance of bird breeding season during site clearance or pre-commencement nesting bird checks	**Not significant**
Loss of habitat for badgers	**Minor Adverse**	Habitat creation proposals throughout GI network to include hedgerows, woodland, scrub and wetland	**Not significant**
Disturbance to badgers during clearance and construction	**Major Adverse**	Adherence to buffer around badger setts Pre-commencement site checks or forthcoming plots	**Not significant**
Loss of habitat for other locally occurring species e.g. otter, water vole and white-clawed crayfish	**Not Significant**		**Not Significant**

Fragmentation and isolation effects on fauna	**Moderate adverse**	Habitat creation proposals throughout GI network to include hedgerows, woodland, scrub and wetland	**Minor Adverse**
Post-completion stage			
Potential effects on habitat through increased recreation/visitor pressure	**Minor adverse**	Green infrastructure management plan to include management of areas for both recreation and biodiversity. Zoning of uses to provide areas of seclusion and more active recreational use.	**Minor adverse**
Potential predation of grass snake by domestic pets	**Minor adverse**	Habitat creation and enhancement proposals throughout GI network to include hedgerows, open grassland, scrub and wetland.	**Not Significant**
Potential predation of farmland and garden birds	**Minor adverse**	Habitat creation proposals throughout GI network to include hedgerows, woodland, scrub and wetland managed for the benefit of farmland and garden birds	**Not Significant**
Disruption of bat dispersal routes through severance and introduction of street lighting	**Minor adverse**	Enhancement of GI habitat to provide a enhanced dispersal and foraging habitat. Sensitive design	**Not significant**
Potential disturbance of badgers and setts	**Minor adverse**	Implementation of buffer zones and/or dense planting to deter access	**Not Significant**
Disruption of dispersal corridors and increased badger road casualties	**Moderate adverse**	Retained hedgerows/features in development Incorporation of badger underpasses where appropriate under main access route at existing field edge. location	**Not Significant**